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The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Temporal-interference (TI) stimulation applies two high-frequency currents whose amplitudes beat at a low difference frequency, a proposal for improving the spatial selectivity of non-invasive electrical stimulation at depth. The superposition carries superposition contains no field component at the beat, so a beat-frequency neural response requires a nonlinearitytheir low difference frequency. We show how neural populations can act as tuned receivers of such inputs: a nonlinearity mixes the two carriers into a difference-frequency component, and recurrent dynamics near a Hopf bifurcation generate and selectively amplify that componentselectively at the network's own rhythm, as $1/\gamma$ in the damping. Generation and amplification are distinct contributions, and a large network response does not by itself mean efficient detection. We compare two modeled detectors. . In a Jansen–Rit columnthe memoryless , the firing-rate sigmoid mixes the carriers through its curvature; in . In an exact quadratic integrate-and-fire mean fieldthe cell's own voltage quadratic mixes the carrier-frequency excursion and the population reads the result through its slope. Both give quadratic scaling in the field, but their coefficients and operating-point dependence differ, and a , the corresponding static curve emerges when the membrane variables relax faster than the input, while the synapses remain dynamic. At faster inputs, mixing depends on the population dynamics and cannot be inferred from the steady-state rate curve does not determine the response to a fast carrier: the crossover from curvature to slope sits at the population 's own frequency, below TI's carriers, and with matched access the two detectors share the carrier roll-off and differ in coefficient and sign. In the local square-law description the generated beat depends on curve alone. Both descriptions generate a leading weak-input beat proportional to the product of the two effective carrier inputs rather than on the envelope excursion, and the network susceptibility sets how strongly it appears in the response. TI and a direct low-frequency drive share the downstream amplifier, so resonance alone gives TI no efficiency advantage. In simulations the beat-locked modulation of the population rate grows toward the bifurcation far effective carrier amplitudes, with membrane filtering counted once. Recurrent dynamics near a Hopf bifurcation amplify the beat as $1/\gamma$, where γ is the modal damping. Connectivity controls this gain independently of the generated drive, and beat-locked rate modulation grows faster than the mean rate, and mean-rate change in the mean field and in a spiking network of 8000 neurons per populationshows the same with weakly locked single cells. The carrier selects the entry point: in an access-filter model a somatic compartment is shut at kilohertz while a fast one stays open, a hypothesis rather than a result. The . Fast cellular access can therefore coexist with slow circuit resonance, although the biological access pathway and the absolute response magnitude remain to be established; the analysis identifies what to measure separately: carrier access, nonlinear beat generation, and circuit tuning. absolute response at human field strengths remain unresolved. The analysis separates access, mixing, and amplification; it also shows why an envelope-only dose map requires a physiological assumption. TI and direct low-frequency stimulation share the downstream amplifier, so resonance alone confers no TI efficiency advantage.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

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1 Introduction

Reaching a deep brain target without stimulating the tissue above or around it is a long-standing goal of non-invasive neuromodulation, and temporal-interference Temporal-interference (TI) stimulation [1] is its most prominent recent proposal. TI was proposed to reach deep brain targets while reducing stimulation of surrounding tissue [1]. It applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs(typically $f_1 \sim 1$, typically with $f_1 \sim 1$–2 kHz , $\Delta f \sim 1$and $\Delta f \sim 1$–100 Hz). Where the two fields overlap, their superposition is an amplitude-modulated (AM) signal : a carrier near f_1 whose envelope beats at the difference frequency with an envelope at Δf. The appeal is spatial: the envelope amplitude peaks in a focal, steerable region at depth, while superficial tissue sees only the (putatively inert) kHz carrier—a promise of non-invasive, focal, deep stimulation that explains the intense interestElectrode currents can steer the region of largest envelope modulation. Whether this also localizes the neural response depends on how neurons transduce the waveform.

A mechanistic question remains: how is a response at the beat frequency generated? Write the field as the sum of the two applied tones,

$$s(t) = \varepsilon_1 \cos(\omega_1 t) + \varepsilon_2 \cos(\omega_2 t), \qquad \omega_{1,2} = 2\pi f_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta f,$$

The difference frequency is absent from the input itself:

$$\mathrm{s(t)} = \varepsilon_1 \cos(\omega_1 \mathrm{t}) + \varepsilon_2 \cos(\omega_2 \mathrm{t}), \qquad \omega_{1,2} = 2\pi \mathrm{f}_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta \mathrm{f}. \tag{1}$$

where Here $\varepsilon_{1,2}$ are the amplitudes of the two tones at the target (refined to post-coupling polarizations effective input amplitudes (related to the applied field in §2.2), $\Delta f = f_2 - f_1$ is the envelope (beat) frequencyΩ is the beat angular frequency, and $f_c = (f_1 + f_2)/2$ the carrier. Its spectrum has support *only* at ω_1 and ω_2 is the carrier frequency. The spectrum contains only the two applied lines (Fig. 3): there is no energy at the envelope frequency Ω. A linear, time-invariant (LTI) system maps a sinusoid to a sinusoid of the same frequency; it system can attenuate or phase-shift the two lines but can *never* synthesize a new line at Ω. The membrane 's passive RC low-pass—often invoked as "the neuron follows the slow envelope"—therefore cannot by itself demodulate TI , a point now widely accepted [2]. Demodulation requires a nonlinearity. This is the founding lesson of AM radio [3]: a receiver cannot recover the message with linear components alone; it needs them but cannot create a new one. Passive membrane filtering therefore cannot demodulate TI [2]; a nonlinear element (a square-law device or a diode) to shift spectral energy down to *baseband*—the must mix the carriers into the low-frequency band near 0 Hz where the message lives—followed by a tuned filter to select the station and a low-pass to read the envelope. band, as in AM reception [3].

Where, then, is the nonlinearity in the brain? The leading biophysical account places the nonlinearity in single-neuron ion channels : Mirzakhalili et al. [2] show Voltage-gated ion channels provide a cellular mixing mechanism in Hodgkin–Huxley and MRG-type mammalian myelinated axon models that TI works through nonlinear, ion-channel-mediated rectification, in which asymmetric voltage-gated conductances rectify the carrier and, after membrane low-pass, follow the envelope—the textbook rectify-then-filter detectorMcIntyre–Richardson–Grill (MRG) myelinated-axon models [2]. Luff et al. [4] directly observed subthreshold difference-frequency oscillations in individual neuronsunder two-tone stimulation, including carriers near 5 kHz. At those carrier frequencies the mixing Mixing persisted during synaptic blockade , whereas but was suppressed by blocking tetrodotoxin-sensitive sodium channelssuppressed it. These recordings provide experimental evidence for cellular mixing alongside the biophysical models. Other studies report that TI thresholds inherit the carrier's frequency dependence, suggesting a mechanism shared with direct kHz stimulation (conduction block , onset response) rather than a dedicated envelope channel [5, 6], and in some peripheral preparations the response is not driven by envelope extraction at all [6]. In vivo, TI reliably alters spike *timing* but not firing rate in the primate brain, and is much weaker . Other preparations show carrier-driven excitation or block rather than envelope extraction [5, 6, 7]. Primate recordings at human-comparable fields report changes in spike timing without a corresponding mean-rate change, with weaker effects than conventional low-frequency stimulation under human-relevant conditions [8]. Recent slice work finds that [8]. In slices, pyramidal and parvalbumin (PV) interneurons respond differentlyand that neurons respond differently, and most pyramidal TI responses disappear when synaptic transmission is blocked, leading Caldas-Martinez et al. to conclude that TI is "largely a network phenomenon" [9]; computational network models point the same way [10, 11], and

current reviews describe the mechanistic account as having moved from passive filtering to ion-channel rectification to under synaptic blockade [9]. Computational studies also find network-mediated inhibition [12]. In peripheral nerves and isolated neurons at high intensity, the kHz carrier itself stimulates and thresholds follow the carrier rather than the envelope [5, 7]responses [10, 11, 12]. These findings distinguish cellular mixing from the network response it can recruit.

We ask which features of the TI response follow from standard population nonlinearities and recurrent dynamics, and where additional biophysics is required. Two ingredients already in standard cortical models carry the argument: a firing-rate *sigmoid*, whose nonzero curvature σ'' makes it a square-law demodulator, and a *second-order-synapse network* whose synaptic kinetics filter the carrier away while a weakly-damped *stable focus* resonantly amplifies the recovered beat component at the network's natural frequency separate carrier access, nonlinear beat generation, and recurrent amplification (Fig. 1). The amplification is a property of the focus nearing criticality—here the Hopf that opens the alpha rhythm—and is not tied to one parameter. (A saddle-node on an invariant circle gives the column a separate, slower rhythm, but through excitability with a diverging period rather than a sharp focus resonance, so its amplification is of a different kind.) Where existing accounts locate the nonlinearity in cellular excitability, we compare two modeled detectors: the coarse-grained population sigmoid, whose curvature detects envelopes for carriers slow compared with the population's own dynamics, and the exact mean field of a quadratic integrate-and-fire population, in which the cell's voltage quadratic mixes a fast carrier and the population supplies the read-out. The two agree on quadratic scaling and disagree on the coefficient and its state dependence. What is common, and what this paper establishes, is nonlinear difference-frequency generation followed by network tuning, carrier-independent at fixed polarization and frequency-selective: a candidate network-level route to TI complementary to the disinhibitory network effect (PV-interneuron silencing) reported by Caldas-Martinez et al. [9]; both place TI's action in the population rather than the single cell. We develop it first in a single first calculation uses a Jansen–Rit column, analyzable in closed form, then in an exact next-generation mean field and the spiking network beneath it, where the nonlinearity is derived rather than assumed; a two-band laminar model (LaNMM), whose own alpha and gamma loops bring it closer to cortex, is treated as an extension firing-rate sigmoid: its curvature mixes the carriers, and a network with second-order synapses amplifies the beat near a Hopf bifurcation. An exact quadratic integrate-and-fire (QIF) mean field connects this static description to cellular dynamics. When membrane relaxation is fast relative to the input, a static population transfer curve emerges while the synapses remain dynamic; otherwise the mixing coefficient is frequency dependent. Simulations in the corresponding spiking network test the population description, and a laminar model extends the calculation to two resonant bands at low carriers.

This work builds on a line of research in which Related neural-mass models implement AM-radio principles *internally*, for the brain's own signals. In our cross-frequency-coupling account of predictive coding [13], the LaNMM realizes a columnar Comparator through sigmoid-enabled signal-to-envelope (SEC) and envelope-to-envelope (EEC) couplings, where error evaluation is literally demodulation—the sign-reversed prediction "un-modulates" the carrier. The Hierarchical Amplitude Modulation (HAM) framework [14] generalizes this to nested envelopes with constant-Q (log-spaced) bands, a candidate origin of $1/f$ spectra—again through the sigmoid's quadratic cross-term. The present work turns this machinery outward: if the cortex is an AM transceiver for its own rhythms, it can also be an AM receiver for an external carrier.

Throughout we distinguish the nonlinear generation of a difference-frequency drive from the network's response to it. The memoryless sigmoid provides the reference calculation; its steady-state curvature need not describe detection at high carrier frequencies, and the exact mean field shows where it fails. The aim is to identify which predictions are shared across the models, which depend on the detector and the input convention, and what additional biophysics connects the results to transcranial stimulationaccounts use the same nonlinear mixing for endogenous cross-frequency coupling and nested amplitude modulation [13, 14]. Here the input is external, making the field-to-cell coupling an additional requirement. The population reduction identifies what follows once the carrier reaches the nonlinearity; it does not identify the anatomical access pathway.

2 Theory

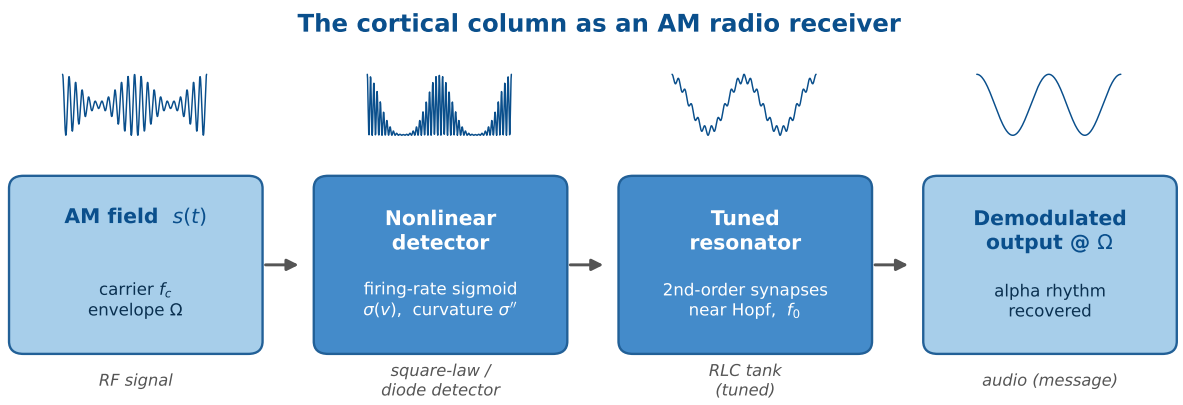


Figure 1: The cortical column as an AM radio receiver (schematic). An AM field (carrier f_c, envelope Ω) reaches a **The cortical column as a tuned receiver.** A nonlinear detector (generates the firing-rate sigmoid, curvature σ''), whose output passes through beat at angular frequency Ω from a tuned resonator (the second-order-synapse network carrier near f_c; a Hopf, recurrent network amplifies it near its natural frequency f_0), recovering a beat-frequency component as an oscillation at Ω. The analogy is functional rather than circuit-identicaldiagram shows the static-sigmoid realization. In a classic conventional AM receiver, the tuned resistor–inductor–capacitor tank sits *before* the detector and circuit selects the carrier before detection [3]; here the sigmoid generates the baseband beat component, and a *baseband* resonator downstream of it selects and amplifies modulation frequencies near the column's own rhythm, so TI "broadcasts" to whichever columns are tuned to its beat frequencylow-frequency output. A The laminar extension also has a carrier-selective stage upstream of the detector does appear in the laminar model (§E).

2.1 Neural mass models Population transfer and the Jansen–Rit circuitslow-synapse limit

A neural mass model (NMM) collapses the activity of a large population ($\sim 10^4$) of similar neurons into a handful of population-averaged describes a population through averaged variables [15, 16]. Two elements recur in the Jansen–Rit-type models considered first. First, each synapse acts as a *linear filter* $\hat{K}$ that converts an incoming In the Jansen–Rit (JR) family, a linear synaptic filter converts presynaptic firing rate $r(t)$ into a postsynaptic potential (PSP) $x(t) = \hat{K}[r]$; equivalently $r = \hat{L}[x]$ with $\hat{L} = \hat{K}^{-1}$ a differential operator whose few coefficients set the synapse's gain, decay time postsynaptic potential $x(t)$:

$$\ddot{x} + 2a\dot{x} + a^2 x = Aa\, r(t),$$

with inverse synaptic time a and gain parameter A; inhibitory synapses use parameters b and delay [15]. In the Jansen–Rit (JR) family the kernel is the "alpha function" $h(t) = Aa\, t\, e^{-at}$ (excitatory; B, b for the inhibitory synapse), i.e. the second-order operator $\hat{L} = (Aa)^{-1}(\mathrm{d}/\mathrm{d}t + a)^2$, so each PSP obeys $\ddot{x} + 2a\dot{x} + a^2 x = Aa\, r(t)$. Second, the PSPs converging on a population sum to a mean membrane potential $v = \sum_s x_s B$. Summed postsynaptic potentials give the mean input v, which a static , saturating *transfer function*—the sigmoid—turns back into a population transfer curve converts to firing rate [17],

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}},$$

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}}. \tag{2}$$

with maximal firing rate Here $2e_0$, half-maximal at the inflection v_0, and slope set by is the maximum rate, v_0 the inflection voltage, and ρ ; it saturates because firing rates are bounded [15]. The sigmoid is the only nonlinearity in the model, and in this model it is what makes (de)modulation possible: it is memoryless, so whatever enters its argument, a kHz carrier included, is mapped to a rate instantaneously. That is an idealization. The curve is the ensemble average of single-neuron input–output relations for slowly varying input, so its curvature σ'' summarizes cellular nonlinearities (thresholds, spike generation, noise) smoothed by the spread of states across the population. Whether a fast carrier reaches that curve unchanged is a question the memoryless model cannot answer; the exact mean field below (Eq. (4)) can, and §2.3 shows that it does not. The network supplies the resonant *amplification* (§2.5). Frequency-selective resonance can also be intrinsic to single

cells, through currents such as I_h and I_M [18]. A companion modeling manuscript by de Palma Aristides et al. [19] reports cell-type- and state-dependent TI frequency selectivity in morphologically detailed hippocampal neurons, with sensitivity to these conductances. Such cellular tuning can coexist with recurrent amplification. The models here isolate the latter: second-order synapses on a loop near criticality generate resonance even when the cells do not [20, 21]. the inverse voltage scale. Nonzero curvature makes this curve a mixer, while its slope sets the local gain of the recurrent circuit.

Assembled into a network, these two elements give the For a synapse from population n to m, the resulting NMM1 transfer-function form of [15] (their Eq. 6.5): for each synapse $m \leftarrow n$,

$$\hat{L}_{m\leftarrow n}[u_{m\leftarrow n}] = C_{m\leftarrow n}\, r_n, \qquad v_m = \Lambda[\hat{E}] + \sum_{n:\, C_{m\leftarrow n}\neq 0} u_{m\leftarrow n}, \qquad r_m = \sigma(v_m),$$

equations are [15]

$$\hat{\mathsf{L}}_{\mathsf{m}\leftarrow\mathsf{n}}[\mathsf{u}_{\mathsf{m}\leftarrow\mathsf{n}}] = \mathsf{C}_{\mathsf{m}\leftarrow\mathsf{n}}\, \mathsf{r}_\mathsf{n}, \qquad \mathsf{v}_\mathsf{m} = \Lambda[\hat{\mathsf{E}}] + \sum_{\mathsf{n}:\, \mathsf{C}_{\mathsf{m}\leftarrow\mathsf{n}}\neq 0} \mathsf{u}_{\mathsf{m}\leftarrow\mathsf{n}}, \qquad \mathsf{r}_\mathsf{m} = \sigma(\mathsf{v}_\mathsf{m}). \tag{3}$$

with $u_{m\leftarrow n}$ the PSP produced at synapse $m \leftarrow n$, The operator $\hat{L}$ is the inverse synaptic filter, $u_{m\leftarrow n}$ its postsynaptic potential, $C_{m\leftarrow n}$ the connectivity, and $\Lambda[\hat{E}]$ any *direct*, non-synaptic perturbation of the membrane. An applied fieldenters precisely here, as the additive term $\Lambda[\hat{E}]$ in the sigmoid's argument, the direct polarization caused by the applied field. This field contribution enters alongside the synaptic PSPs—the starting point for §2.2. The concrete equations and parameters of the models simulated are collected in App. B, in this formalismpotentials, without first passing through a synapse.

This becomes exact one rung up the modeling ladder [15]. For a network of The corresponding population nonlinearity can be derived for all-to-all coupled quadratic integrate-and-fire (QIF) QIF neurons with Lorentzian-distributed excitabilities, the Ott–Antonsen ansatz yields a *closed, exact* mean field for the population . Their exact Montbrió–Pazó–Roxin (MPR) mean field evolves the firing rate r and mean potential voltage v —the next-generation (Montbrió–Pazó–Roxin, MPR) mass [22], completed with the second-order synapses of neural-mass models in the exact "[22]; adding dynamic synapses gives NMM2 " theory [23, 24]:

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t),$$

[23, 24]. In membrane-time units,

$$\dot{\mathsf{r}} = \frac{\Delta}{\pi} + 2\mathsf{rv}, \qquad \dot{\mathsf{v}} = \mathsf{v}^2 + \bar{\eta} - \pi^2\mathsf{r}^2 + \mathsf{I}(\mathsf{t}), \tag{4}$$

with the applied current $I(t)$ entering linearly, the common background excitability where $\bar{\eta}$ (the bifurcation parameter), and the heterogeneity is the mean excitability, $\Delta > 0$ its half-widthΔ setting the smoothing. Here the nonlinearity is *derived, not postulated*: the quadratic v^2 term, inherited from the single-neuron QIF dynamics, mixes whatever the membrane potential carries, and the applied current enters the voltage equation rather than a rate map, and $I(t)$ the normalized input, including recurrent drive when present. Rate and voltage are expressed in the corresponding rescaled units. For a constant input Q the population settles on the rate $r = F(Q)$,

$$F(Q) = \frac{1}{\pi}\sqrt{\frac{Q + \sqrt{Q^2 + \Delta^2}}{2}},$$

total drive $Q = \bar{\eta} + I$, setting both derivatives to zero gives $v = -\Delta/(2\pi r)$ and

$$\mathsf{r} = \mathsf{F}(\mathsf{Q}) = \frac{1}{\pi}\sqrt{\frac{\mathsf{Q} + \sqrt{\mathsf{Q}^2 + \Delta^2}}{2}}. \tag{5}$$

a sigmoid-like curve with a convex foot, This nonlinear curve exists without recurrent coupling. It has an inflection at $Q = \Delta/\sqrt{3}$ and an unbounded concave rise[24]; it is the slow-input limit of the dynamics,

not a logistic. The Jansen–Rit sigmoid is the heuristic (NMM1) stand-in for such a curve. Both models mix two carriers quadratically, but their detector coefficients are not the same object: the memoryless sigmoid's is its curvature, the MPR population's depends on the carrier frequency through the voltage dynamics (§2.3). The same exact theory exhibits weak-field resonance and Arnold tongues that sharpen with self-coupling [24, 23], the near-Hopf amplification we exploit below. , unlike the saturating JR sigmoid.

What the transfer functions represent. In every model the demodulating object is If the population relaxes to this equilibrium faster than synaptic inputs change, NMM2 reduces to a static rate map with dynamic synapses [23, 24]. Restoring the membrane time τ_m and physical firing rate, a *population* input–output map rather than a passive somatic membrane element. In JR/LaNMM it is the sigmoid (2), the effective transfer of an excitable neuronal ensemble (mean PSP in, population firing rate out); in NMM2 the same role is played by the exact MPR quadratic (4). Its curvature is the mesoscopic trace of cellular nonlinearities (thresholds, spike generation, refractoriness, noise, saturation), so for inputs slow compared with the population's own dynamics, reading the demodulation off σ'' is legitimate at this scale; for a fast carrierthe derived v^2 term mixes the voltage excursion and the population curve enters as the read-out gain (§2.3). The membrane low-pass does not contradict this: it bears on *access*—how the carrier reaches the nonlinearity—not on legitimacy.Because an applied field polarizes the membrane directly, in parallel, at every compartment (§2.2), access is set by the entry point, and second-order synapse with time constant τ_s obeys

$$\mathrm{r(t)} \simeq \frac{\mathrm{F(Q(t))}}{\tau_\mathrm{m}}, \qquad \tau_\mathrm{s}^2 \ddot{\mathrm{s}}_\mathrm{syn} + 2\tau_\mathrm{s}\dot{\mathrm{s}}_\mathrm{syn} + \mathrm{s}_\mathrm{syn} = \mathrm{r(t)}, \tag{6}$$

where s_{syn} is the open biophysical question is which fast compartment (axon initial segment, node, or channel) still passes a kHz carrier to that nonlinear transfer (§2.6). Once the carrier is sampled there, filtered rate entering the recurrent drive in Q, with the original coupling normalization retained. This is the neural mass describes the population-level demodulation and near-Hopf amplification of the recovered beat componentslow-synapse limit: shortening the effective membrane time makes the transfer appear static while the network retains its synaptic dynamics. The relevant separation is between the population relaxation time τ_{rel} and τ_s. Applying the same static map directly to a carrier of angular frequency ω_c requires the stronger condition $\omega_c\tau_{\mathrm{rel}} \ll 1$. A population can therefore be quasistatic for its synaptic inputs yet dynamic for a kHz carrier.

The transfer curve summarizes cellular nonlinearities only within this regime. The exact QIF reduction includes its spike-generating quadratic and excitability distribution; it does not by itself reproduce the sodium-channel kinetics implicated by Luff et al. [4]. Those mechanisms can enter an effective population description through a calibrated dynamic input–output functional. A steady-state rate curve alone does not specify its frequency dependence. Section 2.3 calculates this dependence for QIF and recovers Eq. (5) in the adiabatic limit.

Cellular and circuit resonance are also distinct. Conductances such as I_h and I_M can tune single neurons [18]; the companion manuscript of de Palma Aristides et al. [19] reports cell-type- and state-dependent TI frequency selectivity in hippocampal neurons, sensitive to these conductances. Recurrent excitatory–inhibitory circuits can generate resonance without intrinsic cellular resonance [20, 21]. In the fast-membrane limit, any retained resonance belongs to the remaining synaptic circuit; modes requiring finite membrane dynamics need not persist.

The single JR column couples a pyramidalpopulation to one excitatory and one inhibitory interneuron population through this synapse+sigmoid motif, with the field entering pyramidal, excitatory-interneuron, and inhibitory-interneuron populations, with field input to the pyramidal sigmoid as $s(t) = \Lambda[\hat{E}]$ (§2.2) and the and local field potential (LFP) proxy $v = y_1 - y_2$. Its equations (Eqs. (28)) and parameters (Table 5) are collected in (App. B.

For the biophysical realization (§E)we use a second model built from the same ingredients: the laminar neural mass model). The laminar extension (LaNMM) [13, 25] , which embeds this synapse+sigmoid motif in a two-band cortical column—a deep Jansen–Rit-like subcircuit (adds coupled alpha and gamma circuits: deep pyramidal $P_1$with , spiny-stellate (SS), and somatostatin (SST) interneurons; intrinsic alpha, ∼10 Hz) coupled to a superficial PING-like (pyramidal–interneuron network gamma) subcircuit (populations, and a superficial pyramidal $P_2$with PV interneurons; intrinsic gamma, ∼40 Hz). The single JR column lets us isolate and analyze the mechanism; the LaNMM then

exhibits it in a column with its own fast/slow loops and two native resonances. Everything below applies to both: the field enters the sigmoid argument as a membrane polarization, and the recovered beat component drives a resonant circuit/PV circuit. These models retain dynamic synapses and static rate curves; NMM2 and its spiking realization retain membrane dynamics as well.

2.2 From applied field to a squared-envelope beat component

We model the applied TI use the quasistatic electromagnetic approximation for the applied field $E(t)$ $(\mathrm{V/m})$as a quasi-static perturbation of the membrane potential that the spike-generating nonlinearity reads. "Quasi-static" means we neglect electromagnetic , neglecting propagation and induction(wavelengths $\gg$ head, frequencies far below tissue relaxation), so the field acts as an instantaneouspolarization added inside the firing-rate map, . This does not make membrane charging instantaneous. Let $s(t)$ be the polarization after the cellular access response; the static population model reads

$$r(t) = \sigma\big(v(t) + s(t)\big), \qquad E(t) = E_1 \cos \omega_1 t + E_2 \cos \omega_2 t. \tag{7}$$

Unlike synaptic drive, an applied field does not enter through the synaptic kernel: it polarizes the membrane directly and simultaneously at every compartment—soma, dendriteThe field acts directly on soma, dendrites, axon initial segment (AIS), terminal—each and terminals, each with its own coupling and its own access filter. The perturbation $s(t)$ reaching the nonlinearity is therefore a sum over these frequency response. In a reduced description, their contributions to the effective input are represented by parallel entry points, or "doors",

$$s(t) = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}(t), \qquad \lambda_f^c = \lambda_0^c \, |H_c(\omega_c)|, \tag{8}$$

where λ_0^c is the DC (quasi-static) coupling of door static polarization length of compartment c—the weak-field polarization length $\delta V = \boldsymbol{\lambda}\cdot\mathbf{E}$ of transcranial-stimulation modeling [15, 26]—and λ_f^c is its effective coupling at the carrier, the DC value rolled off by that door's own filter H_c. Both are limits of one compartment equation , $C\dot{\mathbf{V}} + A\mathbf{V} = \mathbf{I}^0 - \mathbf{p}\,E$ [23] , whose response to an alternating field is $\delta\mathbf{V}(\omega) = -(A + i\omega C)^{-1}\mathbf{p}\,E(\omega)$: the quasi-static λ_0 is its value at $\omega \to 0$ and H_c H_c its access filter, and $\omega_c = 2\pi f_c$ [15, 26]. Equation (8) uses amplitude gains; relative phases can be retained in the complex transfer. The passive compartment equation $C\dot{\mathbf{V}} + A\mathbf{V} = \mathbf{I}^0 - \mathbf{p}E$ [23] gives $\delta\mathbf{V}(\omega) = -(A + i\omega C)^{-1}\mathbf{p}E(\omega)$, with capacitance matrix C, conductance matrix A, endogenous current $\mathbf{I}^0$, and field-to-current coupling $\mathbf{p}$. The static gain and its frequency dependence belong to this one response. A model driven by post-coupling polarization and a model driven by current through its own membrane equation must therefore count that filtering once. Prescribing $\lambda_0 E$ inside a static sigmoid and injecting the current $\mathbf{p}\,E$ into a model that carries its own $C\dot{V}$, as the exact mean field does (§2.3), are the two ends of this one chain; the membrane filter is applied onceeither way, and a comparison of the two detectors is only meaningful with the field inserted at the same point. The field is thus not only a somatic membrane perturbation, and which door dominates is set by the carrier: at DC the soma/dendrite has the largest coupling (the whole-cell dipole), but its filter cuts it at kHz while a fast door stays open, so $\lambda_f^{\mathrm{soma}}$ collapses and the dominant term crosses over (§2.6). The amplitudes

For a dominant entry point, $\varepsilon_j = \lambda_f(\omega_j)E_j$ denotes the polarization reaching the nonlinearityare the post-coupling polarizations $\varepsilon_j = \lambda_f E_j$ of the open door (door index suppressed when only the open door matters), which is what Eq. (1) contains rather than the raw applied field. The main text takes the favorable direct limit in which a fast door (axonal or nodal) samples the carrier before appreciable attenuation, $\lambda_f \approx \lambda_0$, isolating the sigmoid+resonance principle. The direct-coupling calculation takes $\lambda_f \approx \lambda_0$ to isolate mixing and recurrent amplification; §2.6 restores the roll-off and shows the carrier dependence entering quadratically, $A_\Omega \propto |H_c(f_c)|^2$frequency-dependent access. A fast compartment could justify this approximation, but its identity and coupling are not inferred from the population model.

Expanding σ about an operating point v^* (a Volterra/Taylor expansion, as in [14]) gives $\sigma(v^*+s) \approx \sigma(v^*) + \sigma'(v^*)\,s + \frac{1}{2}\sigma''(v^*)\,s^2 + \dots$ The truncation holds only while $|s|$ is small against the sigmoid's voltage scale $1/\rho \approx 1.8$ mV . Human TI polarizations are $\sim 10^{-2}$ mV, over two orders below that, so the quadratic term is the leadingnonlinearity throughout the regime this paper treats. Rodent TI is not in that regime: the fields are $\sim 10^3$ times larger, the sigmoid saturates, and the expansion fails. §5 returns to what that costs when rodent results are scaled to humans.

Intuitively, the difficulty is that the field carries power only near the carrier ω_c, far from the slow envelope frequency Ω, so some operation must move power down to Ω. The *linear* term cannot: it passes the input lines straight through, reproducing the carrier and its two flanking *sidebands* (the lines at $\omega_c \pm \Omega$ created by the modulation), but still nothing at Ω. The *quadratic* term can: squaring the field multiplies the fast carrier by itself, and because $\cos^2$ has a constant (zero-frequency) part, the squaring folds power down to *baseband*—the low-frequency band near 0 Hz, where the slow envelope lives. This is *rectification*, what a diode does in an AM radio. The requires $|s| \ll 1/\rho \approx 1.8$ mV and a nonzero second derivative for quadratic mixing to be leading. Larger excursions require the full transfer curve. The linear term preserves the two input frequencies. The quadratic cross term generates their difference through the identity $2\cos\omega_1 t\cos\omega_2 t = \cos\Omega t + \cos(\omega_1+\omega_2)t$shows where the beat comes from: the cross term of s^2. Averaging over the carrier (the fast $\cos^2\omega_j t$ terms average to $\frac{1}{2}$, the sum-frequency term to zero),

$$\langle\sigma(v^*+s)\rangle_c = \sigma(v^*) + \tfrac{1}{4}\sigma''(v^*)\,(\varepsilon_1^2+\varepsilon_2^2) + \tfrac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2\cos\Omega t + O(\varepsilon^4), \tag{9}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon_1\varepsilon_2, \tag{10}$$

where $\simeq$ means to leading order in the polarizations, here and in what follows. The same expansion leaves a constant (DC) term, $\frac{1}{4}\sigma''(\varepsilon_1^2+\varepsilon_2^2)$, which any read-out of the mean rate must include.

What the quadratic term recovers is not the envelope but its square. Writing $A(t)$ for the instantaneous envelope of the two-tone field, $A^2(t) = \varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos\Omega t$, the low-pass part of the squared signal is exactly

$$\mathrm{LPF}\big[s^2(t)\big] = \tfrac{1}{2}A^2(t) = \frac{\varepsilon_1^2+\varepsilon_2^2}{2} + \varepsilon_1\varepsilon_2\cos\Omega t. \tag{11}$$

This is square-law detection in the strict sense: the population transduces A^2, not A. The distinction matters because A^2 is exactly sinusoidal at Ω whereas A is not—at a matched target ($\varepsilon_1=\varepsilon_2=\varepsilon$), $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine.

The amplitude-modulated surrogate. The dense perturbative sweeps below use the three-line waveform

$$s(t) = \varepsilon\,[\,1+m\cos\Omega t\,]\cos\omega_c t, \qquad \omega_c = 2\pi f_c, \tag{12}$$

with modulation depth m, because it parameterizes a sweep by one amplitude. It is not the physical signal: it carries a line at ω_c that the two-tone sum lacks. The two agree at Ω to leading order (the surrogate gives $A_\Omega = \frac{1}{2}\sigma''\varepsilon^2 m$) and are *not* equivalent beyond it. For matched genuine tones $\mathrm{LPF}[s^2] = \varepsilon^2(1+\cos\Omega t)$, whereas for the $m=1$ AM surrogate $\mathrm{LPF}[s^2] = \varepsilon^2(\frac{3}{4}+\cos\Omega t + \frac{1}{4}\cos 2\Omega t)$: the same Ω amplitude, but a different DC term and a spurious second harmonic at 25% of the fundamental. Since that second harmonic is precisely the observable that distinguishes transduction classes (§2.4), every claim that turns on the waveform is run with genuine two tones: the demodulation demonstration (Fig. 3), the curvature and square-law controls (Fig. 6), the laminar maps (Figs. S1, S7) and the kHz demonstration. The AM surrogate is retained for the dense perturbative sweeps in (Ω, p) and (Ω, C), which read only the leading Ω component, and Fig. 6d verifies that the two agree there.

In the language of [14], the quadratic cross-term $s_1 s_2$ between a slow and a fast input (the second-order Volterra kernel) implements amplitude modulation/demodulation and the associated intermodulation lines. Demodulation is thus itself an intermodulation product: the line at Ω is the *difference frequency* of the two applied tones (ω_1 and ω_2), generated by the same quadratic term that—acting a second time, now that, acting again on an intrinsic rhythm ω_0—produces , produces the $\omega_0 \pm \Omega$ sidebands seen at the spiking level (Fig. S12b). One nonlinear mixer, applied in cascade. Equation (10) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier

ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \lambda_f E_0$, so the field required for a given effect rises with f_c (§2.6)—consistent with the experimental consensus. Carrier independence is thus a statement about the recovered beat component at fixed polarization, not about the applied field.

2.3 Static transfer versus dynamic mixing: the The exact mean field connects static and dynamic mixing

The sigmoid of Eq. (10) is memoryless, so the carrier traverses the curve and the curvature is the detector at any carrier frequency. A population of neurons is not memoryless: its rate relaxes and rings on its own timescales, and an applied current polarizes the membrane before any spike-generating nonlinearity sees it. The exact mean field makes the consequence explicit. Take a single adiabatic condition in §2.1 determines whether the static curve describes carrier mixing. Consider an uncoupled MPR population (Eq. (4), $\tau = 1$, no coupling)with constant input membrane time set to one), with constant drive Q and a weak current carrier added current $a\cos\omega t$added to the voltage equation. Expanding to . To second order in $a$and averaging over the carrier, the rate acquires a constant shift $a^2\,C(Q,\omega)$ with

$$C(Q,\omega) = \frac{r^*\,[\omega^2 + 3d^2 - \nu^2]}{(d^2+\nu^2)\{[d^2+\nu^2-\omega^2]^2 + 4d^2\omega^2\}}, \qquad r^* = F(Q), \quad d = \frac{\Delta}{\pi r^*}, \quad \nu = 2\pi r^*,$$

where , the carrier-averaged rate shift is $a^2 C(Q,\omega)$, where

$$\mathsf{C}(\mathsf{Q},\omega) = \frac{\mathsf{r}^*\,[\omega^2 + 3\mathsf{d}^2 - \nu^2]}{(\mathsf{d}^2+\nu^2)\{[\mathsf{d}^2+\nu^2-\omega^2]^2 + 4\mathsf{d}^2\omega^2\}}, \qquad \mathsf{r}^* = \mathsf{F}(\mathsf{Q}), \quad \mathsf{d} = \frac{\Delta}{\pi \mathsf{r}^*}, \quad \nu = 2\pi \mathsf{r}^*. \tag{13}$$

The fixed-point eigenvalues are $-d \pm i\nu$: d is the population's relaxation rate and ν its ringing frequency (the linearization about the fixed point has eigenvalues $-d \pm i\nu$). Its two limits are the two detectors,

$$C(Q,\omega) \xrightarrow[\omega\to 0]{} \tfrac{1}{4}F''(Q), \qquad C(Q,\omega) \xrightarrow[\omega\gg\nu]{} \frac{F'(Q)}{2\omega^2}.$$

the ringing angular frequency, both in membrane-time units. The limiting responses are

$$\mathsf{C}(\mathsf{Q},\omega) \xrightarrow[\omega\to 0]{} \tfrac{1}{4}\mathsf{F}''(\mathsf{Q}), \qquad \mathsf{C}(\mathsf{Q},\omega) \xrightarrow[\omega\gg\mathsf{d},\nu]{} \frac{\mathsf{F}'(\mathsf{Q})}{2\omega^2}. \tag{14}$$

For a slow carrier the In the first limit the population tracks its steady-state curve and the static curvature rectifies, as in the Jansen–Rit calculation. For a fast carrier the voltage responds as an integrator, with an excursion $\sim a/\omega$ and a quarter-cycle lag; the v^2 term mixes that excursion into a constant, and the population converts the constant to a rate through its *slope*. The coefficient is then positive whatever the operating point and largest at the inflection, where F'' vanishes. static curve, whose curvature rectifies the input. In the second, the voltage excursion is approximately a/ω; its square supplies a slow drive that the population reads through F'. Above the inflection the two of F, these limits have opposite signs and the coefficient changes sign vanishes at $\omega_z = \sqrt{\nu^2 - 3d^2}$ (Fig. 7a). The crossover sits at the population's own frequency, tens of hertz for the gamma-generating PING of §4.5; the 300 Hz carrier used there, and TI's kHz carriers , and kHz carriers are on the fast side . for the PING parameters used below. That classification is parameter-dependent: a shorter effective membrane time can bring the same physical carrier into the adiabatic regime.

The two limits are not two carrier penalties. The linear response of the voltage to the current Membrane access is already included in this calculation. The linear voltage response is $H_v(\omega) = (i\omega + d)/[(i\omega + d)^2 + \nu^2]$, the population's own membrane: state-shaped below ν, the bare capacitance $1/(i\omega)$ above it. Dividing C by $|H_v|^2$ gives the detector . Dividing by its squared magnitude gives the mixing coefficient at fixed voltage excursion,

$$K(\omega) = \frac{C(Q,\omega)}{|H_v(\omega)|^2} = \frac{F'(Q)}{2}\,\frac{\omega^2 + 3d^2 - \nu^2}{\omega^2 + d^2},$$

$$\mathsf{K}(\omega) = \frac{\mathsf{C}(\mathsf{Q},\omega)}{|\mathsf{H}_\mathsf{v}(\omega)|^2} = \frac{\mathsf{F}'(\mathsf{Q})}{2}\,\frac{\omega^2 + 3\mathsf{d}^2 - \nu^2}{\omega^2 + \mathsf{d}^2}. \tag{15}$$

which is bounded: it runs from the static curvature per unit squared excursion at $\omega \to 0$ This coefficient tends to $F'/2$ for $\omega \gg \nu$, vanishes at ω_z, and carries no power of ω above the population's rates. All of the at high frequency. The $1/\omega^2$ suppression at fixed current sits in comes from $|H_v|^2$: it is the membrane term $\tau\dot{v}$ integrating the injected current, the same low-pass that , the membrane filtering already represented by Eq. (8)places in front of the sigmoid, here computed by the model rather than assumed. Since $F'' = 4r^*(3d^2 - \nu^2)/(d^2 + \nu^2)^3$, the inflection of F is the point $\nu^2 = 3d^2$, and above it the fast coefficient $F'/2$ and the static curvature have opposite signs. Relative to a static curve read on ; it is not an additional detector filter. A static curve placed after the same membrane , the dynamics therefore multiply the fast-carrier coefficient by $d^2/(3d^2 - \nu^2)$, a constant set by the operating point (0.35, 0.66 and -0.03 at $r^* \approx 10$, 17 and 37 Hz for $\tau = 15$ ms, $\Delta = 1$), not a further roll-off. With matched access the memoryless and the dynamic detector share the leading carrier scaling; the detector sets the prefactor, its sign and its structure near the population's own frequencieshas the same leading carrier roll-off, although its mixing coefficient and sign can differ. The steady-state curvature is $F'' = 4r^*(3d^2 - \nu^2)/(d^2 + \nu^2)^3$, so the inflection occurs at $\nu^2 = 3d^2$.

The same reasoning fixes what a two-tone field generates in For the full eight-state PING of (Eq. (27). In the fast-carrier limit each tone's voltage phasor is), each high-frequency current tone of amplitude a_j produces a voltage phasor $a_j/(i\omega_j\tau_E)$, and the . The v_E^2 term produces therefore supplies a difference-frequency source of magnitude $a_1 a_2/(\tau_E^2\omega_1\omega_2)$ in the τ_E-divided voltage equation , independent of the operating point; the response is that source passed inside the equation for $\tau_E\dot{v}_E$. After division by τ_E, this source is propagated through the network 's susceptibility at Ω. Against the exact second-order response (full quadratic source and full resolvent) at At four stable operating points, the fast-carrier source is resulting response agrees with the full second-order calculation to within 1.1% at 300 Hz, 0.1% at 1 kHz, and 0.03% at 2 kHz (Fig. 7b). Two consequences carry through the paper. Agreement in the ε^2 scaling between the sigmoid and

The carrier and beat have different routes to the firing rate. At high frequency, the exact mean field does not mean the same detector: the coefficients and their operating-point dependence differ, and the sign reversal of Eq. (10) at the inflection is a property of the memoryless model. And carrier independence at fixed polarization survives in the dynamic model, with polarization now the carrier-frequency voltage excursion $a_j/(\omega_j\tau_E)$: at fixed current the generated source rolls off as $1/(\omega_1\omega_2)$ from the membrane dynamics themselves. The rate's own linear response falls faster, as $1/\omega^2$: the membrane integrates the drive into the voltage voltage equation integrates the input once, and the rate equation integrates the voltage again, which is what a dynamic transfer function shows when it is driven at the carriergiving a linear rate response proportional to $1/\omega^2$. The beat does not inherit the second power, because the product is formed is generated in the voltage equation , before the rate integrates: the second-order susceptibility at the difference frequency is $\propto 1/(\omega_1\omega_2)$ while the first-order one is $\propto 1/\omega^2$. In a single population driven by tones at 995 before this second integration and 1005 Hz ($\tau = 15$ ms, $r^* \approx 10$ Hz) the carrier line in the rate is 0.02 of what a static curve on the same membrane would give, while the beat line is 0.32 of it, both to four digits as this chain predicts. Table 1 collects the exponents. Relative to a static curve on the same membrane, the dynamic population supplies one additional high-frequency integration , from voltage to rate; it acts on the carrier in the rate and not on the beat, which is processed through the population is read through the susceptibility at Ω, a factor that carries no carrier-frequency scaling. Of the three quadratic terms of . Thus its carrier dependence is $1/(\omega_1\omega_2)$, not the product of two linear rate responses. Among the quadratic terms in Eq. (4), v^2 mixes two voltage excursions at $1/(\omega_1\omega_2)$, $2rv$ a rate and a voltage excursion at $1/\omega^3$, and $\pi^2 r^2$ two rate excursions at $1/\omega^4$, so v^2 controls the fast-carrier beatcontributes at order ω^{-2}, rv at ω^{-3}, and r^2 at ω^{-4} for nearby fast tones. Table 1 summarizes the comparison.

2.4 Transduction class and its spatial and spectral signatures

A neural response proportional to the squared envelope is not equivalent to one proportional to the envelope. A detector implements some transduction $F(A)$ of the local envelope $A(t)$, and the two laws predict different amplitude scaling, harmonic content and spatial weighting, all three measurable. This section says how to tell them apart. .

To make "class" concrete, compute what a single site emits at baseband. For a narrowband field $s(t) = A(t)\cos\omega_c t$ passing through a memoryless nonlinearity g, the low-frequency (baseband) output Y_{LF} is the

stage	static curve, same membrane	dynamic population
current → voltage	ω^{-1}	ω^{-1}
voltage → rate, carrier	1	ω^{-1}
current → rate, carrier	ω^{-1}	ω^{-2}
two tones → beat in the rate	$(\omega_1\omega_2)^{-1}$	$(\omega_1\omega_2)^{-1}\times$ const.

Table 1: Fast-carrier exponents of the two detectors on the same scaling with matched membrane access, at fixed beat frequency and a nonzero leading mixing coefficient. The dynamic population suppresses integrates the carrier in the rate once more than between voltage and rate; the static curve does, but the beat is formed in the voltage equation before that integrationstep. The constant in the last row is $d^2/(3d^2-\nu^2)$ times the population susceptibility at Ω: it depends Prefactors depend on the operating point , and can change sign, but not on the carrierbeat-frequency susceptibility.

average of g over one carrier cycle, $Y_{\mathrm{LF}}(A) = (2\pi)^{-1}\int_0^{2\pi} g(v^* + A\cos\theta)\,\mathrm{d}\theta$. :

$$\mathsf{Y}_{\mathrm{LF}}(\mathsf{A}) = (2\pi)^{-1}\int_0^{2\pi} \mathsf{g}(\mathsf{v}^* + \mathsf{A}\cos\theta)\,\mathrm{d}\theta.$$

Odd powers of A average away, leaving only even ones:

$$Y_{\mathrm{LF}}(A) = g(v^*) + \tfrac{1}{4}g''(v^*)A^2 + \tfrac{1}{64}g^{(4)}(v^*)A^4 + \cdots \tag{16}$$

A weak, smooth nonlinearity therefore transduces A^2 generically, whatever its detailed shape—which is what the sigmoid mechanism of §2.2 does, and why Eq. (10) is quadratic. Recovering A itself requires a qualitatively different detector: strong rectification. A full-wave rectifier gives $\mathrm{LPF}[\,|A\cos\omega_c t|\,] = (2/\pi)A$ and a half-wave rectifier $(1/\pi)A$, both linear in the envelope, as does a diode-and-capacitor peak detector. A hard threshold is not one of them: a step detector with offset δ transduces $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and nothing below it, which is neither linear nor quadratic in A (App. D). Asymmetric channel rectification, threshold crossing and release events are candidate biological realizations of rectifiers of one kind or another. Under a linear downstream response and fixed coupling, the amplitude sweep below distinguishes a quadratic effective drive from one proportional to the envelope; the derivation of §2.2 assumes a smooth transfer, and this section asks what would show that assumption to fail.

The two classes are distinguishable at a single site. At a matched target, $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos u| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1}\frac{(-1)^{n+1}}{4n^2-1}\cos 2nu$ gives a harmonic ladder in ratios $A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}A^2 = \varepsilon^2(1+\cos\Omega t)$ contains only DC and Ω. For a mixed detector $F(A) = c_1A + c_2A^2$ the pre-network ratio is

$$r_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_\Omega}\right| = \frac{1/5}{1 + \frac{3\pi}{4}\,c_2\varepsilon/c_1}, \tag{17}$$

so measuring r_{det} estimates $c_2\varepsilon/c_1$ directly.

Two effects make the harmonic ratio usable only with care, and the second decides how to run the test. First, a smooth detector does generate a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, so the square-law signature is a *vanishing* ratio in the weak-field limit, not an exact zero. Second, and more seriously, the recurrent network multiplies each harmonic by its own transfer gain, $r_{\mathrm{meas}} = r_{\mathrm{det}}\,|G(2\Omega)/G(\Omega)|$. A resonant column suppresses the second harmonic hard: driven at $\Omega = \omega_0$ with $\mathcal{Q} = 10$, a true 20% ratio reads as 0.7%, enough to make an envelope follower look square-law. The resonance is thus precisely what would hide the discriminator, so the test must be run *off* resonance: at $\Delta f \simeq f_0/4$ both lines sit on the flat part of G, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ across $\mathcal{Q} = 3$–30, or with G measured independently by a weak sweep of direct low-frequency stimulation (transcranial alternating-current stimulation, tACS) and deconvolved.

An amplitude sweep tests the scaling rather than one ratio. An envelope-following detector gives $Y_\Omega \propto \varepsilon$ with the harmonic ratio pinned at 1/5; a weak square-law detector gives $Y_\Omega \propto \varepsilon^2$ with a ratio growing as ε^2. The log–log slope of the envelope-locked response against field amplitude therefore separates the classes without requiring absolute calibration; the ε^2 control of §4.4 is that measurement, performed in the model.

Beyond harmonics and amplitude scaling, the transduction class fixes the spatial map. Define the excursion as $\mathcal{M}[y] \equiv \max_t y - \min_t y$ of a periodic quantity $y(t)$. The conventional TI envelope map is $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$, whereas the square-law detector responds to $\mathcal{M}[\frac{1}{2}A^2] = 2\varepsilon_1\varepsilon_2$, so the two are related pointwise by

$$\mathcal{M}\left[\tfrac{1}{2}A^2\right] = \max(\varepsilon_1, \varepsilon_2)\, \mathcal{M}[A]. \tag{18}$$

The square-law map is the conventional envelope-excursion map multiplied, at each point, by the dominant local polarization. It therefore does not preserve conventional TI focality. In typical montages, where the dominant carrier is substantially larger superficially than at depth, this reweighting degrades deep-to-superficial contrast: at $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially against $(0.5, 0.5)$ at a matched deep target, $\mathcal{M}[A]$ favors the superficial point by $2\times$ and $\mathcal{M}[\frac{1}{2}A^2]$ by $40\times$. The magnitude of the degradation is not fixed by this algebra and must be evaluated from the actual vector field distribution; a sufficiently weak second carrier can suppress the superficial product instead. The product law is not an account of TI focality.

The selectivity this mechanism does supply is dynamical rather than spatial. Writing $R_\Omega(\mathbf{x})$ for the demodulated response at the envelope frequency Ω produced at location $\mathbf{x}$, and $G_\mathbf{x}(\Omega)$ for the local network susceptibility there,

$$R_\Omega(\mathbf{x}) \;\propto\; \sigma''(v^*_\mathbf{x})\, G_\mathbf{x}(\Omega)\, \varepsilon_1(\mathbf{x})\, \varepsilon_2(\mathbf{x}), \tag{19}$$

a product of three factors. Field overlap sets where a difference-frequency drive is generated at all; the operating point and the network susceptibility set which of the driven populations amplifies it. Resonance does not override an unfavorable field product, but it can compensate for one. This is the radio analogy taken literally: received signal strength times receiver tuning.

Equation (19) is written for scalar projected fields. In general the coupling is vectorial and compartment-specific, $\varepsilon_i(\mathbf{x}) = \boldsymbol{\lambda}\cdot\mathbf{E}_i(\mathbf{x})$, so the mixing product $(\boldsymbol{\lambda}\cdot\mathbf{E}_1)(\boldsymbol{\lambda}\cdot\mathbf{E}_2)$ is signed: its magnitude follows the product of the projected amplitudes, while its phase inverts with either the sign of σ'' or the relative sign of the two projections. The derivations below take the collinear, scalar case.

2.5 Amplification: the synaptic low-pass and the network resonance

The recovered beat component at Ω survives the loop only if the carrier is filtered out of it, and is then amplified by the network. For the carrier to leave, it need only be high enough that the synaptic low-pass suppresses it relative to the envelope, and "inert"—it must not itself fall in, or beatinto, the network's responsive band, nor engage carrier-specific mechanisms (the ion-channel/kHz pathway, §2.6). Within those bounds the response is flat in f_c (Fig. 5, left). The synaptic suppression is concrete: each JR synapse applies the operator $\hat{L}_s$ with impulse response $h(t) = Aa\,t\,e^{-at}$, i.e. Synaptic filtering suppresses the carrier after the nonlinearity has generated the beat. Each JR excitatory synapse has transfer function $H_s(\omega) = Aa/(i\omega + a)^2$, a critically-damped second-order low-pass with corner $a/2\pi \approx 16$characteristic frequency $a/(2\pi) \approx 16$ Hz. It passes the ∼10 Hz envelope ($|H_s(\omega)|/|H_s(0)| \sim 0.7$) while suppressing a 10 Hz component with relative amplitude about 0.7, while reducing a 100 Hz carrier to ∼2.5% component to about 2.5% (Fig. 5, right), so the carrier is filtered out of the loop *after* the sigmoid has used it to demodulate. Thus two distinct mechanisms shape the closed-loop dynamics: the sigmoid generates a squared-envelope component at the beat frequency, whose amplitude is inherently independent of the carrier, and the synaptic low-pass then removes the carrier itself from the spectrum). Provided the carrier does not itself excite a responsive mode, the loop therefore reads predominantly the generated low-frequency drive.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives a leading eigenvalue pair $\lambda_\pm = -\gamma \pm i\omega_0$; near the

Hopf the response is dominated by this critical mode, so the whole network behaves as a single weakly damped harmonic oscillator with damping γ and natural frequency ω_0 (the gain below is its one-mode, center-manifold reduction). Just below Below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (10) at Ω gives the textbook resonance gain

$$G(\Omega) \propto \left[(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0}, \tag{20}$$

so the steady response $\sim A_\Omega\, G(\Omega)$ peaks at $\Omega = \omega_0$ (a Lorentzian in power) with a height that diverges as the Hopf is approached, within the weak-drive regime where the linear response holds. Here ω_0 is the imaginary part of the eigenvalue; the undamped frequency of the canonical oscillator differs from it by a term of order γ^2, immaterial near the Hopf. Just above Above the Hopf the mass is an autonomous limit cycle at ω_0, and the demodulated drive *entrains* it (an Arnold tongue), giving a response peak near ω_0 that grows with cycle amplitude.

Collecting the stages, the response at the beat is

$$R_\Omega \simeq G(\Omega;\gamma)\; D\; \lambda_f(\omega_1) E_1\, \lambda_f(\omega_2) E_2, \tag{21}$$

with $\lambda_f(\omega_c) = \lambda_0 |H_c(\omega_c)|$ the access of the open door (Eq. (8)), D the detector coefficient ($\frac{1}{2}\sigma''(v^*)$ for the memoryless sigmoid; for the exact mean field the slope-read coefficient where λ_f maps field to local carrier polarization, D converts the product of those polarizations into a beat drive, and G maps that drive to the observed response. Their normalization depends on where the generated drive enters. For JR, $D = \frac{1}{2}\sigma''(v^*)$ and G is the response to a generated rate drive. For fast-carrier QIF in the normalized units of Eq. (15), constant above the population's own rates) (4), $D = 1$ for the v^2 source and G the network gain of Eq. (20). The carrier frequency enters only through λ_f, the beat frequency only through G, and the operating point through includes its conversion through the voltage and rate dynamics. Thus the population readout is included once. In these limits D and $\gamma$is independent of carrier frequency at fixed polarization; across the dynamic crossover it acquires frequency dependence. A carrier sweep and a beat sweep therefore measure different stages , and the two detectors differ in D, not in the carrier dependenceconstrain different stages rather than a single effective gain.

The resonance amplifies but cannot detect. The universal does not supply the quadratic mixer. The cubic Hopf normal form , whose nonlinearity is odd, responds to an in-band drive yet produces no line at Ω at any order in the field but has no second-order difference-frequency source (App. C), so detection and amplification are separate requirements rather than two aspects of proximity to criticality. . A large resonant susceptibility alone therefore does not establish beat generation.

2.6 Access and carrier independence: fast elements and Carrier access depends on the kHz carriereffective membrane response

Carrier independence (at fixed polarization) holds for the direct coupling (7); the access filter H_c of Eq. (8) restores the carrier dependence, and *which* entry point is open sets its corner. Take first the *somatic door*: the membrane (and dendritic cable) low-passes the field before the spike nonlinearity. (The demodulating σ'' is the *population* transfer; the door filter is the upstream *access* stage—this section asks only whether the carrier survives to it.) Modeling the somatic door as a first-order filter does not imply independence at fixed applied field. To quantify access, consider a first-order membrane response $H_m(\omega) = 1/(1+i\omega\tau_m)$, the carrier amplitude reaching the detector is $\varepsilon_{\rm eff}(f_c) = \varepsilon/\sqrt{1+(\omega_c\tau_m)^2}$ (that is, $\lambda_f^{\rm soma} E_0$ in the notation of Eq. (8)), and since $\Omega \ll \omega_c$ the three AM lines are attenuated almost equally, so

$$A_\Omega(f_c) \;\simeq\; \tfrac{1}{2}\,\sigma''(v^*)\, m\, \varepsilon_{\rm eff}^2 \;=\; \frac{A_\Omega(0)}{1+(\omega_c\tau_m)^2} \;\xrightarrow[\omega_c\tau_m \gg 1]{}\; A_\Omega(0)\,\frac{1}{(\omega_c\tau_m)^2}.$$

, with effective time constant τ_m. For nearby carriers, the two polarizations acquire nearly the same attenuation. The quadratic beat then satisfies

$$\frac{A_\Omega(f_c)}{A_\Omega^{\rm direct}} \simeq |H_m(\omega_c)|^2 = \frac{1}{1+(\omega_c\tau_m)^2} \sim \frac{1}{(\omega_c\tau_m)^2} \quad (\omega_c\tau_m \gg 1), \tag{22}$$

The demodulated response rolls off as $1/f_c^2$ above the membrane corner (Fig. S5). For a pyramidal soma/dendrite the passive time constant is $\tau_m \sim 10$where A_Ω^{direct} is the beat generated with the same static coupling and no access attenuation. This filter belongs either in the field-to-polarization map or in the membrane equation that computes that polarization.

Somatic passive time constants are commonly 10–30 ms(human L2/3 pyramidal cells: ∼12; reported human pyramidal values include approximately 12 ms [27] and 16.5 ± 3.7 ms [28], which bracket the ∼16 ms we adopt for the soma, corner ∼10 Hz), so the carrier-power . For the illustrative $\tau_m = 16$ ms used here, the power factor $|H_m|^2$ is only $\sim 10^{-4}$ at 1about 10^{-4} at 1 kHz, $\sim 2.5 \times 10^{-5}$ at 2 kHzand $\sim 10^{-6}$ at 10$2.5 \times 10^{-5}$ at 2 kHz, and 10^{-6} at 10 kHz. Passive somatic demodulation is therefore negligible at TI's kHz carriers—a four- to six-order-of-magnitude suppression. Equation (22) is the response computed through the somatic door: correct, but the door a kHz carrier finds shut. A static nonlinearity placed after this somatic response therefore receives strongly attenuated carriers.

In this access-filter model, at kHz the somatic door is shut and a faster one is open. A fast element (axon initial segment (AIS), node of Ranvier; $\tau \approx 0.2$ ms , effective corner ∼800 Hz) low-passes the same field only mildly ($|H|^2 \approx 0.39$ at 1 kHz , A shorter effective time changes this conclusion quantitatively. The sensitivity case $\tau_m = 0.2$ ms gives $|H_m|^2 \approx 0.39$ at 1 kHz and 0.14 at 2 kHz , still $\sim 6 \times 10^{-3}$ at 10 kHz; 2 kHz (Fig. S5). This is not the parameter swap $\tau_m \to \tau_{\text{fast}}$ within one door but a distinct entry point of These are assumed access parameters, not measured TI parameters of the AIS or a node. They can represent a faster compartment in Eq. (8), in parallel with the soma rather than in series with it. A square-law nonlinearity there produces a real envelope-frequency current; because that component is generated at Ω *after* the access filter, no upstream pole can remove it, and it is exactly what the slow network reads. The per-neuron modulation is small, but two factors make it count: the envelope sits in the band the membrane passes, and the population near a Hopf amplifies it by $G \sim 1/\gamma$ (or motivate a shorter membrane time in a QIF reduction. In the latter description, the approach to a static transfer is governed by Eq. (20)). Demodulation at fast elements, resonantly amplified at the population, is thus a candidate route for TI at realistic carriers, consistent with ion-channel rectification at the node [2] but adding network resonance; our direct-coupling demonstration is the $\tau \to 0$ idealization of this limit, and a direct kHz-carrier run (Fig. S9) confirms it. (6); the membrane filter must not be applied again outside the QIF equations. A short access time improves carrier transmission without, by itself, specifying the mixing coefficient or the remaining circuit resonance.

Entry point (door) Element	Input	Access Illustrative corner	Dominates when Interpretation
synapse → dendrite → soma JR excitatory synapse	rate r	∼16 Hz	normal operation downstream synaptic filter
field → soma / dendrite somatic access	field ε E	∼10 Hz	DC, slow tES τ_m = 16 ms example
field → AIS fast effective access	field ε E	∼**800 Hz** 800 Hz	**kHz carriers (TI)** assumed $\tau_m = 0.2$ ms
Na_v activation gate (within σ) channel kinetics	— local voltage	∼1 kHz model-dependent	sets the ceiling part of the dynamic detector

Table 2: Entry points into the sigmoid argument of EqDistinct response stages. (8). The same field enters through several doors; access corners are order-of-magnitude, and which door dominates is set by the carrier. The AIS door (bold) is the leading candidate for TIillustrate specified time constants; the compartment is fast row does not yet settledidentify an anatomical compartment. Presynaptic terminals polarize strongly butSynaptic filtering follows beat generation, as isopotential compartments, are no faster than whereas access controls the soma (§5)carrier reaching the nonlinearity.

The somatic and fast doors can be told apart only at kHz. At low frequency the AIS and soma are electrotonically one compartment, so "somatic" and "spike-initiating" potential are the same number and the two doors are indistinguishable: the neural-mass state variable is degenerate at DC. The degeneracy lifts as the AC space constant $\ell(\omega)$ contracts from its DC

valueℓ_0,

$$\ell(\omega) = \ell_0 \, (1 + \omega^2 \tau_m^2)^{-1/4},$$

Geometry can also alter the relevant response mode. For a passive cable, the magnitude of the frequency-dependent space constant is

$$\ell(\omega) = \ell_0 (1 + \omega^2 \tau_{\mathrm{m}}^2)^{-1/4}, \tag{23}$$

by roughly tenfold at 1 kHz for where ℓ_0 is its static value. At 1 kHz and $\tau_m = 16$ ms—to tens of μm, comparable to the AIS length itself. A kHz carrier is thus the regime in which the demodulating compartment becomes determinate; note that $\varepsilon_{\mathrm{eff}}$ as defined above absorbs only the membrane pole, not this contractionms, this length is about one tenth of its static value, so compartments that are strongly coupled at low frequency can respond differently to the carrier. A somatic charging time alone therefore does not determine every extracellular-field response mode. The periaxonal and field-charging evidence is considered in §5.

One lumped assumption then remains in the Jansen–Rit description: that σ is static; the exact mean field removes it (§2.3). A static σ is the $\tau_{\mathrm{gate}} \to 0$ idealization of a spike-initiation nonlinearity whose $\mathrm{Na_v}$ activation gate has $\tau \approx 0.1$–0.2 ms, a corner near 1 kHz; above it σ'' itself rolls off by a factor $|H_{\mathrm{gate}}(\omega_c)|^2$. This gate, not the membrane, sets the frequency ceiling of the mechanism, and it is measurable: a carrier sweep at fixed envelope frequency and field should roll off only in the kHz range, where the access corner ($\sim$800 Hz) and the gate corner ($\sim$1 kHz) combine, not at the $\sim$10 Hz corner of the somatic door.

Because transcranial electrical stimulation (tES)—and TI, a tES with a kHz carrier—is weak and *modulatory*, the fast-element route is a small, sustained bias of the most excitable compartments, not "spiking the AIS at the beat frequency." Since the applied field adds to endogenous drive, a weak bias near threshold shifts spike probability and timing through stochastic resonance [29, 30]. The population sigmoid (2) is itself a mean-field stochastic-resonance device—its smooth slope is the spread of thresholds and noise across the population—so the bias shifts the mean rate through σ' while Channel kinetics can introduce further frequency dependence when they are not already represented in the detector. An explicit gate filter H_{gate} would require its own dynamical model or measurement; a static sigmoid does not imply a universal 1 kHz gate cutoff. In particular, the demodulated beat component emerges through the curvature σ''QIF coefficient in Eq. (13) includes the dynamics of that reduced model, not a separately calibrated sodium-channel response. Carrier sweeps can constrain the combined access and mixing response, while beat-frequency sweeps probe the downstream amplifier.

Both halves are state-dependent, detection through the operating point and amplification through $1/\gamma$, so TI efficacy should track brain state, not the stimulus alone. The 100 Hz carrier used throughout is not merely convenient: the membrane attenuation is mild even for moderately fast elements ($|H_m|^2 \approx 0.98$ at $\tau = 0.2$ ms, $\sim$0.1 at $\tau = 5$ ms), so a protocol with $f_c \sim 100$ Hz and Δf swept through a region's band is a clean near-term test. The trade-off is spatial: a 100 Hz carrier is less inert off-target than a kHz one (whose subthreshold carrier is the source of TI's focality), so 100 Hz probes the mechanism while kHz optimizes selectivity.

2.7 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 6). The Population Excitation–Inhibition Index (PEIX) of [13] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\mathrm{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \mathrm{sech}^2\Big[\tfrac{\rho}{2}\,(v^* - v_0)\Big] \in (0, 1], \tag{24}$$

a dimensionless read-out of where v^* sits, the side given by $\mathrm{sign}(v^* - v_0)$. PEIX thus tracks σ' (maximal at the inflection v_0), whereas the demodulation gain tracks σ'' (zero at v_0, peaked on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks exactly where the demodulation gain vanishes. This complementarity is a property of the memoryless detector. In the exact mean field at a fast carrier the rectified DC term follows the slope F' and is largest at the inflection (§2.3), so the relation between PEIX and TI efficacy is model-dependent; PEIX

remains a measurable read-out of the operating point. In the Jansen–Rit detector, anything shifting v^* (neuromodulation, attention, arousal, or pathology such as psychedelics or PV-interneuron dysfunction in Alzheimer's disease) scales, and can flip the sign of, the TI response; in the dynamic detector the same shifts act through F' and the susceptibility, without a sign rule. The state dependence is the falsifiable handle; the sign reversal is specific to the memoryless model.

3 Methods

3.1 Single-column Jansen–Rit: integration, lock-in, and controls

(All symbols and units are collected in App. A, Table 4.) We integrate the single-column JR model of §2.1 (Eqs. (28), parameters in Table 5) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (28), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at Ω in v is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S9 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and 1.5×10^{-5} s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.6 are analytic evaluations of Eq. (22), not simulations. For each measurement the system is settled for 10–16 s (longer near the Hopf, where damping is weak) before a 3–20 s measurement window; the resonance sweeps use a 20 s window so the lock-in averages the limit-cycle beating on the entrainment side. Parameter sweeps are vectorized: the state is an $(N, 6)$ array advanced simultaneously for all N parameter combinations, so a full resonance map runs in seconds. The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically and diagonalizing it: the Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\lambda_+$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \mathrm{Im}\,\lambda_+/2\pi \approx 11.1$ Hz—consistent with the continuation and with the JR bifurcation analysis of [31]. For $p > 315$ the fixed point is a stable focus (resonator); between the saddle-node on an invariant circle (SNIC, $p \approx 114$) and the Hopf the column free-runs on the alpha limit cycle.

The response at the envelope frequency is read out with a software lock-in amplifier referenced to Ω—a standard way to isolate the component of a signal at one chosen frequency: multiply by sine and cosine at Ω and average, so everything *not* at Ω cancels and only the envelope-locked amplitude survives. Concretely we multiply $v(t)$ by quadrature references and integrate over a Hann-windowed measurement window to obtain the in-phase (I) and quadrature (Q) components,

$$I = \frac{2}{\sum_k w_k} \sum_k w_k \,(v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k \,(v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2}, \quad (25)$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to -A\sin\phi$, so $A_\Omega \to A$exactly—the ; the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. The one subtlety is that out-of-band components must not Out-of-band components can leak into the estimate; the dangerous one is the , particularly the DC level of v ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). Over an exact integer number of Ω-periods $\sum_k w_k \cos \Omega t_k = 0$ and there is no leakage, but in general it is nonzero, so we subtract $\bar{v}$ explicitly; this makes the metric exact for a tone and robust accurate down to the 10^{-8} level needed for the kHz analysis (§2.6). The signed in-phase component I alone recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 6).

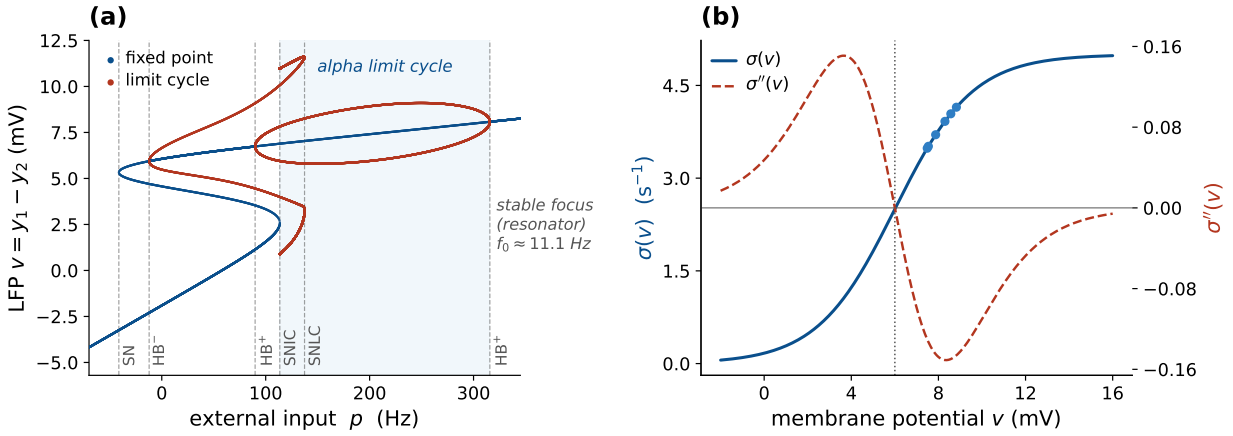


Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature. *(a)* JR bifurcation diagram from numerical continuation—the LFP $v = y_1 - y_2$ along the fixed-point branch (blue) and the min/max of the limit-cycle branches (red) vs. external input p, with *solid* stable and *faded* unstable. The alpha limit cycle (shaded) is bounded by a saddle-node-on-invariant-circle (SNIC, $p \approx 114$) and the supercritical Hopf ($p \approx 315$, $f_0 \approx 11.1$ Hz); beyond the Hopf the fixed point is a stable focus—the near-critical resonator regime used throughout. The column is a weakly-damped focusnear *both* bifurcations, so it resonates on either approach—more slowly near the SNICnear-Hopf resonator used here. Further codim-1 points are marked (saddle-node SN, Hopf $HB^{\pm}$, saddle-node of limit cycles SNLC). *(b)* The sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection v_0. Dots mark the operating points reached by the closed-loop model for the p values used here.

Three auxiliary constructions support the main results. Throughout we contrast the *open-loop detector*—the We compare the *open-loop detector*, the firing-rate sigmoid alone at a fixed operating point v^* , with no recurrence—against without recurrence, with the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the two-tone field through σ at v^* and applies Eq. (25), isolating the pure demodulation law (Figs. 5, 6, S5). The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in Eq. (28) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against finite differences. All code is openly available (see *Code and data availability*).

3.2 Laminar two-band model (LaNMM)

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [13, 25], with the field entering exactly as in the single JR column—as column, as a membrane polarization added inside the sigmoid argument of the target population,

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\}, \tag{26}$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude ε in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S2). Coupling the field this way, rather than as a A perturbation of an external input *rate* , matters for more than tidiness: a rate cannot go negative, so a rate-coupled AM drive must be clipped would require clipping at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at Δf into the drive before the population nonlinearity ever saw it. A signed membrane polarization carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, ε is directly comparable between the two models. Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid simultaneously, as in the JR sweeps. In place of the lock-in (Eq. (25)) we read the alpha-band

(8–12 Hz) power of the P_1 and P_2 membrane potentials as the mean squared magnitude of the Hann-windowed FFT over the band, in arbitrary units, over a post-transient window, a phase-agnostic measure suited to the entrainment regime, where lock-in phase slips would otherwise complicate the read-out. Two sweeps probe the column: an amplitude–beat sweep over $(\varepsilon, \Delta f)$ at fixed carrier (the Arnold tongue), and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed ε, built on the LaNMM module of [13]. The maps (Figs. S1, S7) show the *signed* stimulation-induced change, alpha power minus the unstimulated ($\varepsilon = 0$) baseline, on a zero-centered scale; over most of the P_1-driven grids the change is negative. For the alpha resonance we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (25)) of v_{P_1} at Δf.

3.3 Exact next-generation mean field (NMM2 PING)

We test the mechanism in the exact next-generation mean field (NMM2, §2.1), following the Rosetta Stone formulation of [15] (which places the Montbrió–Pazó–Roxin reduction [22] on a common footing with the JR and LaNMM models used here), where the firing-rate nonlinearity is the derived MPR dynamics of Eq. (4) rather than a postulated sigmoid. We use a two-population excitatory–inhibitory (PING) motif [23, 24, 15]: each population $\alpha \in \{E, I\}$ is an MPR mean field (firing rate r_α, mean potential v_α) with membrane time τ_α, coupled through second-order synapses,

$$\begin{aligned} \tau_E \dot{r}_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, & \tau_E \dot{v}_E &= v_E^2 + \bar{\eta} - (\pi\tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + I_E(t), \\ \tau_I \dot{r}_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, & \tau_I \dot{v}_I &= v_I^2 + \bar{\eta} - (\pi\tau_I r_I)^2 + \tau_I C\,(s_E - 2 s_I), \end{aligned} \tag{27}$$

with second-order synaptic activations $s_E = \hat{K}_a[r_E]$ (excitatory, time constant τ_a) and $s_I = \hat{K}_g[r_I]$ (inhibitory, τ_g), each obeying the same critically-damped operator as the JR synapse, $\tau^2 \ddot{s} + 2\tau \dot{s} + s = r$; here C is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows—i.e. rows: the E membrane sees $C\,(s_E - s_I)$ and the I membrane $C\,(s_E - 2s_I)$, the only asymmetry being the doubled $I \to I$ weight $A_{ii} = 2$. The common background drive $\bar{\eta}$ —the same in both populations, with no separate input current—is is the bifurcation parameter: raising it past a supercritical Hopf at $\bar{\eta}_H = 1.110$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory voltage equation as a current-like forcing, $I_E(t) = P\,E(t)$ in the convention of the exact mean field [23], with the dipole coupling P instantaneous, as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) of amplitude A_f in the dimensionless MPR units of v_E. The E membrane $\tau_E \dot{v}_E$ integrates this forcing into the carrier-frequency excursion $A_f/(\omega\tau_E)$ that the exact quadratic v_E^2 term rectifies, with no postulated sigmoid; the PING as run is therefore the somatic door of Eq. (8) computes membrane access with $\tau_m = \tau_E$, computed rather than assumed. We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (25)) of r_E at Δf over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [15]): $\Delta = 1$, global coupling $C = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz), located by bisection on the leading eigenvalue of the mean-field Jacobian. The simulations retain these finite membrane times; Eq. (6) characterizes the fast-membrane limit.

Two parameters reach this gamma Hopf, and we use both as criticality knobs—just as the external drive p and the coupling C both reach the JR alpha Hopf. The background drive $\bar{\eta}$ (above) is swept for the resonance map (Fig. S8); the global coupling $C\,(\equiv J)$ is swept for the J-curve (Fig. S10): we vary J at fixed $\bar{\eta}$ toward the gamma Hopf J^* (located by bisection on the leading Jacobian eigenvalue) and read the lock-in peak of r_E over Δf. Unlike the JR J-curve, *no operating-point pinning is needed*: at this carrier the generated difference-frequency source is operating-point independent to about 1% (§2.3), so sweeping J changes the damping γ and the modal projections but not the source, and the gain is read as the response per unit source.

3.4 QIF spiking network (microscale of NMM2)

NMM2 is the exact mean field of an all-to-all network of QIF neurons [22, 15, 24], so we integrate that network directly to test the prediction at the spike level (§4.8): $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (27)), excitabilities $\eta_j \sim$ Lorentzian$(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout. The bifurcation parameter is the common background drive $\bar{\eta}$, with the gamma Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz) from the mean-field Jacobian; the spiking network shows the same threshold and an autonomous $f_0 \approx 53$ Hz, with no finite-size offset to explain. The carrier is $f_c = 300$ Hz and the field amplitude $A_f = 35$. Because this is a *gamma* circuit, the TI beat Δf is set in the gamma band: $\Delta f = 55$ Hz *near* the forced gamma resonance ($\bar{\eta} = 0.85$, below the Hopf) and $\Delta f = 42$ Hz in the oscillatory regime ($\bar{\eta} = 1.5$, above the Hopf)—a detuning of $\approx$11 Hz below the autonomous $f_0 \approx 53$ Hz that places the beat outside the 1:1 Arnold tongue, so the above-Hopf response is gating, not entrainment (§F). The network is integrated with Euler steps of 0.0025 ms, settled for 800 ms (over four relaxation times at $\bar{\eta} = 0.85$, where the mean-field relaxation time is 176 ms) and measured over an integer number of beat cycles, 44 at 55 Hz and 34 at 42 Hz (800 and 810 ms), so that the finite-window bias of phase statistics vanishes. Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega / \langle r_E \rangle$ (a stable measure; the off→on lock-in ratio would divide by the field-free floor), and per neuron by the pairwise phase consistency of each neuron's spike phases at Δf [32], which is unbiased in the spike count, over neurons with at least ten spikes in the window. Halving the time step changes the lock-in and the pooled vector strength by under 1%; doubling the spike-detection threshold changes the lock-in by about 1% and the mean rate by 2%, but the pooled vector strength in the forced regime by about 45%, so phase concentration, unlike rate modulation, is sensitive to how spike times are defined.

3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The coupling sweeps of Figs. 9 and S11, and the transcranial alternating-current stimulation (tACS) control of Fig. S3, use the single JR column with the operating point v^* pinned: for each global connectivity C we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds v^* (hence $\sigma''(v^*)$) fixed, vary $C\,(= J)$ toward the alpha Hopf, and read γ, ω_0 from the 6×6 Jacobian. The field is the AM surrogate of Eq. (12) for TI (Fig. 9; AC lock-in vs DC mean rate for Fig. S11) or a direct in-band sinusoid $s(t) = \varepsilon \cos \Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S3), each read out by the lock-in of v at Δf.

3.6 Vocabulary: forced response, entrainment, and gating

"Entrainment" is used in several senses, so we fix three terms. *Forced response* is the driven, non-autonomous case below the Hopf, quantified by the lock-in amplitude at the beat Δf (Eq. (25)). *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator [33]; in the maps we screen for it with the dominance criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency, a necessary but not sufficient condition for capture. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [34, 35]. Capture and gating are dynamically different, and the same word would otherwise cover both; Appendix F gives the metrics and the Arnold-tongue protocol.

4 Results

Two rungs of the modeling ladder carry the evidence: the single Jansen–Rit column (closed form, memoryless detector) and the exact next-generation mean field (NMM2, the nonlinearity derived

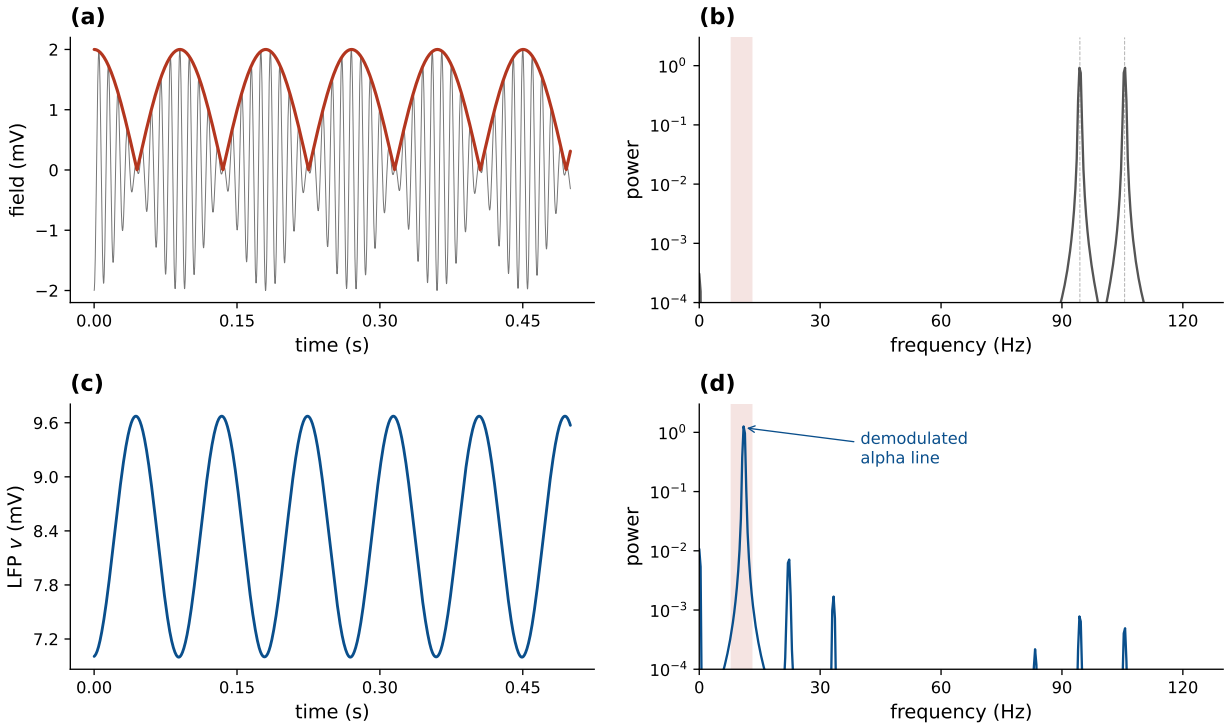


Figure 3: An alpha rhythm is synthesized from a two-tone TI input (Jansen–Rit below Hopf). The field is the physical TI waveform, two equal tones at $f_{1,2} = f_c \mp \Delta f/2$, not the amplitude-modulated surrogate. *(a)* The field $s(t)$ and its envelope $A(t) = 2\varepsilon|\cos(\Omega t/2)|$—a rectified cosine, not a sinusoid; it is A^2 that is sinusoidal at Ω (§2.2). *(b)* The input amplitude spectrum has lines only at f_1 and f_2 (94.45 and 105.55 Hz, $\Delta f = 11.1$ Hz) and no third line at the carrier; in the alpha band (shaded) it is 1.6×10^{-8} of the tone amplitude, finite-window leakage rather than physical power. *(c)* The output $v = y_1 - y_2$ oscillates at the envelope frequency. *(d)* The output spectrum has a strong peak at the beat, 11.1 Hz, synthesized by the sigmoid and absent from the input.

rather than postulated) with its spiking network. Each shows the same pair, quadratic difference-frequency generation followed by near-Hopf amplification, with different detector coefficients. Two mesoscale signatures follow: the network's own coupling sets the gain at fixed generated drive (the J-curve, §4.6), and the resonance amplifies the envelope-locked modulation far more than the mean rate (§4.7), which the QIF network tests at the spike level. The laminar two-band column, at carriers of 25–80 Hz and drives up to 3 mV, is an extension (App. E).

4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

The system is oriented by Fig. 2: a clean supercritical Hopf at $p \approx 315$ with $f_0 \approx 11.1$ Hz, and a sigmoid whose curvature peaks well off the inflection, so the closed-loop operating points sit where σ'' is large. The core result is Fig. 3: an input with no spectral power in the alpha band (lines only at f_c and $f_c \pm \Delta f$) produces a clean ∼11 Hz output oscillation locked to the envelope, with a strong alpha peak in the output spectrum absent from the input.

4.2 Near the Hopf, the demodulated beat drives a sharp resonance

The resonance is quantified in Fig. 4. In the fixed-point regime ($p > 315$, a stable focus) the response is a Lorentzian centered at f_0 whose height grows monotonically as the Hopf is approached ($0.23 \to 0.43 \to 0.70$ mV; Fig. 4), i.e. gain $\propto 1/\gamma$. This is genuine forced gain: with no drive the focus is silent at Ω, so all power at Ω is synthesized by the demodulating nonlinearity and amplified by the recurrent loop. Above the Hopf ($p < 315$) the column free-runs at f_0 and the demodulated drive instead phase-locks that autonomous alpha within a 1:1 Arnold tongue; because lock-in

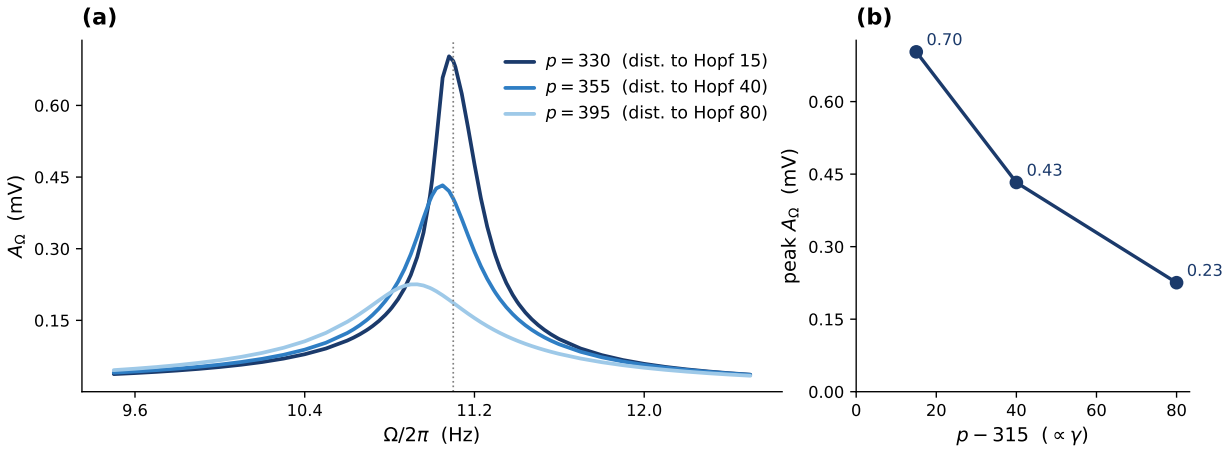


Figure 4: The demodulated beat drives a resonance that sharpens toward the Hopf (Jansen–Rit). Fixed-point regime ($p > 315$, stable focus): lock-in response at Ω vs. envelope frequency ($\varepsilon = 0.3$ mV). *(a)* The demodulated drive forces a Lorentzian resonance at $f_0 \approx 11.1$ Hz whose peak redshifts slightly as the damping grows ($\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\gamma^2}$; $11.08 \to 10.93$ Hz across the three curves); the response carries no power at Ω in the absence of drive, so this is genuine forced gain. *(b)* The peak height grows as the Hopf is approached ($p \to 315$, distance to Hopf $\to 0$). Three points cannot establish a power law; the $1/\gamma$ scaling is quantified over a decade, together with its saturation, in Fig. 9(c). The complementary above-Hopf regime, where the demodulated drive *entrains* the autonomous alpha within a 1:1 Arnold tongue, is characterized with the appropriate observable (output frequency f_{out}, not lock-in amplitude) in Fig. S4.

amplitude at Ω then conflates the driven response with the free-running cycle, we characterize that regime with the output frequency f_{out} rather than the lock-in amplitude (Fig. S4).

4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 5). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner (Fig. S5): passive somatic demodulation is suppressed four to six orders of magnitude at kHz , whereas a fast element ($\tau \approx 0.2$ ms) retains an appreciable response into the kHz bandin the 16 ms access model, whereas the assumed 0.2 ms response retains appreciable carrier access.

To pre-empt the objection that the demonstration runs only at 100 Hz, we We also drive the full recurrent column with a genuine kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant τ) for each coupling route before it reaches the sigmoid. The column demodulates and resonates at the recovered alpha well into the kHz band when the nonlinearity sits at a fast element ($\tau \approx 0.2$in the fast-access case ($\tau = 0.2$ ms), exactly as Eq. (22) predicts (Fig. S9a). Driving the fast-element route with two *genuine* tones at $f_{1,2} = 1994.54$ and 2005.46 Hz, centered near 2 kHz and beating at 10.91 Hz, the input carries 2.2×10^{-13} of its spectral amplitude in the 8–14 Hz band, i.e. none, while the output carries a line at the beat (Fig. S6), with the attenuation set by the analytic $|H_m|^2 = 0.137$ at 2 kHz for $\tau = 0.2$ ms. The kHz claim therefore rests on the physical waveform , not on an extrapolation from the AM surrogate. That the bare unmodulated kHz carrier is itself weakly effective—the premise of TI's focality—is consistent with the limited sensitivity of hippocampal network oscillations two-tone waveform under the specified access assumption. The anatomical route is not determined by this calculation. Limited sensitivity to unmodulated kHz fields has been reported in hippocampal network oscillations [36].

4.4 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 6)—demodulation is the sigmoid-curvature term of Eq. (10), and the closed-loop operating points sit where σ'' is large and negative.

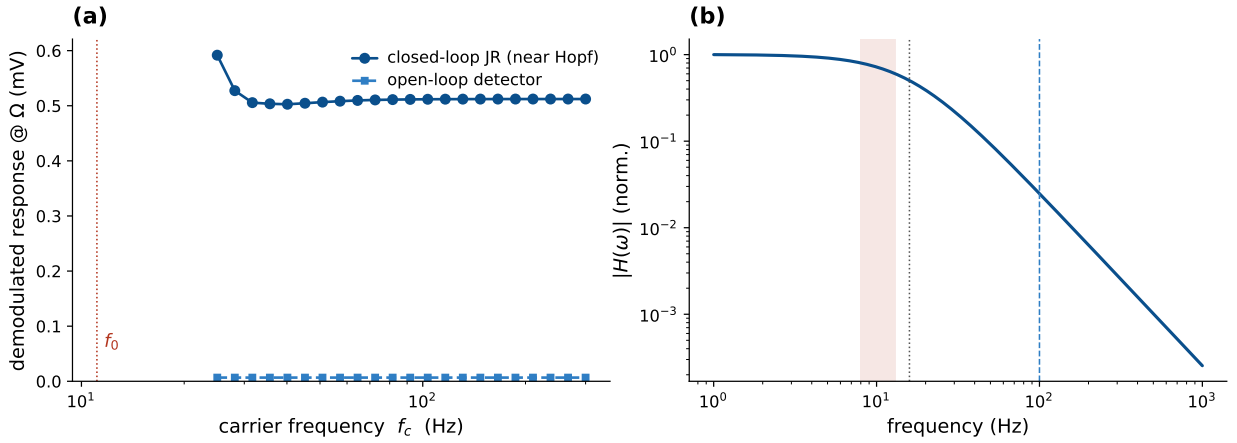


Figure 5: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is flat across carrier frequency $f_c = 25$–300 Hz, both for the open-loop detector (exactly, by Eq. (10)) and for the closed-loop JR near the Hopf (which adds ∼60× resonant gain); deviations appear only as f_c approaches the natural band. *(b)* The second-order-synapse low-pass $|H_s(\omega)|/|H_s(0)|$ passes the envelope (∼0.7) and suppresses the 100 Hz carrier (∼2.5%). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

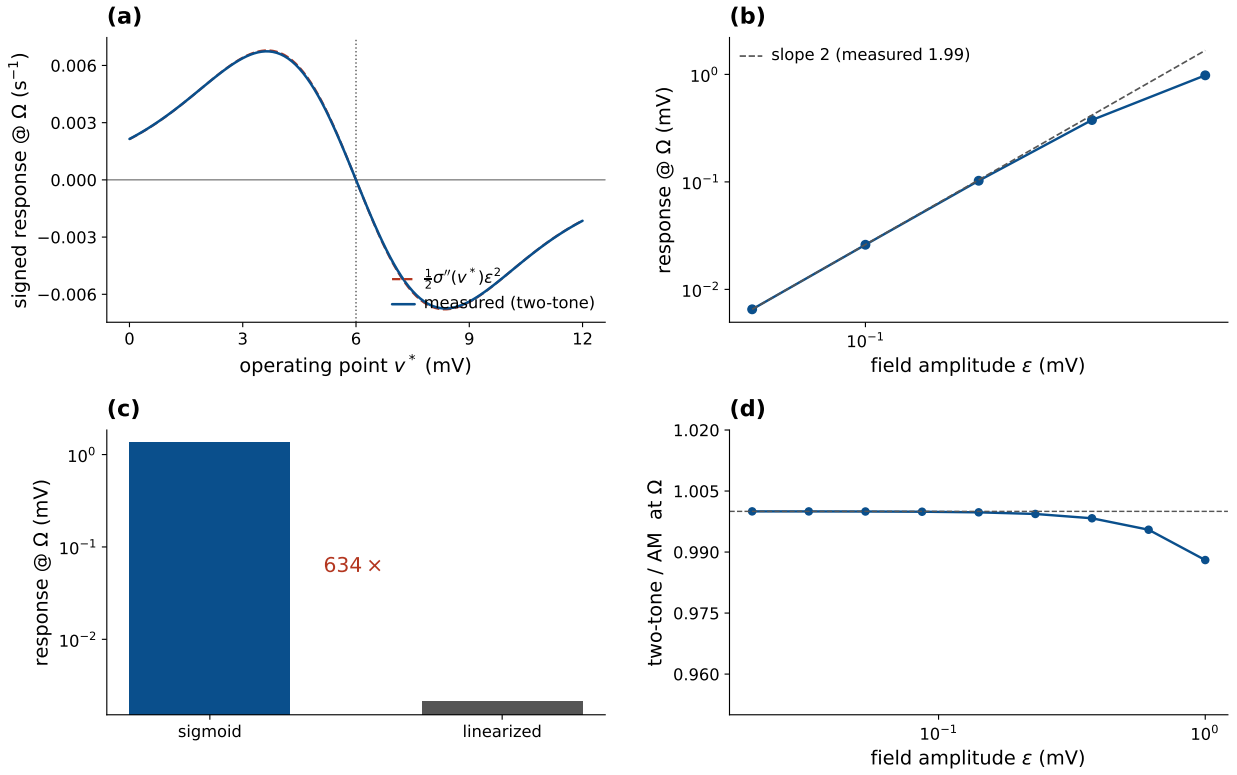


Figure 6: The curvature law, the square law, and the surrogate control (Jansen–Rit, genuine two tones). *(a)* The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2$ across the operating range (maximum deviation 0.9%), vanishing at the inflection v_0 and reversing sign across it. The mechanism is sign-agnostic: σ'' sets sign and phase while $|\sigma''|$ sets gain, so the resonance picture is unchanged up to an overall sign in the $\sigma'' > 0$ lobe, which this sweep spans. The sign flip is itself a prediction (§4.9). *(b)* The closed-loop response scales as ε^2, measured slope 1.99. *(c)* Replacing the field-receiving sigmoid by its tangent collapses the response 634×. *(d)* Surrogate control: the two-tone/AM ratio at Ω approaches 1 as $\varepsilon \to 0$ (1.0000 at 0.02 mV, 0.988 at 1 mV), licensing the AM surrogate for the dense sweeps of Fig. 9, which read only that component, and for nothing else.

Two controls test the proposed source of the beat component (Fig. 6b,c): the response scales as ε^2 at small drive (measured log–log slope 1.99 against the predicted 2), and replacing the field-receiving sigmoid by its linearization collapses the response by 634× (1.34 → 0.0021 mV)—the nonlinearity is the detector, and its curvature is the gain. A third panel checks the surrogate at one operating point: for the open-loop detector at $v^* = 8$ mV the two-tone and AM responses at Ω agree to within 10^{-4} at $\varepsilon = 0.02$ mV and to 1.2% at 1 mV (Fig. 6d). On that basis the dense perturbative sweeps below, which read only the Ω component, use the surrogate; the comparison does not test the recurrent response near the Hopf, where the two waveforms' different DC terms could shift the operating state.

4.5 Exact mean-field confirmation: the NMM2 PING gamma generator

The JR column and the LaNMM both use the heuristic firing-rate sigmoid (NMM1). As a third, independent test we move to the exact next-generation mean field (NMM2, Eq. (4)), where the demodulating nonlinearity is the *derived* quadratic v^2 term rather than a postulated curve. We drive the two-population E–I PING motif of Eq. (27) (Methods)—a gamma generator that , which loses stability through a supercritical Hopf (at $\bar{\eta}_H = 1.110$, $f_0 = 54.0$ Hz), the gamma counterpart of the JR alpha Hopf—with the , with the two-tone field entering the excitatory current, so the carrier is rectified by the exact v_E^2 term.

Driving with a two-tone TI field (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) and locking in the excitatory rate r_E at Δf reproduces the whole picture generation and amplification of the beat (Figs. 8, S8). Below the Hopf the response is a forced resonance at the gamma frequency whose height grows as the bifurcation is approached—the approached, following the $1/\gamma$ gain of Eq. (20); above it the demodulated drive entrains the autonomous gamma. The two-dimensional map traces a ridge at $\Delta f \approx f_0$ that intensifies through the Hopf. Unlike the static-sigmoid models, the resonant frequency itself *shifts with the operating point*—the ridge is diagonal, not vertical—a direct signature of the dynamic, state-dependent transfer of the exact theory Compared with the JR sweep shown here, the NMM2 resonant frequency shifts appreciably with the operating point, producing a diagonal ridge [24]. That the beat is generated and resonantly amplified with the nonlinearity *derived* from QIF spiking dynamics, rather than assumed, shows the receiver behavior is not an artifact of the heuristic sigmoid. The detector coefficient, however, is not the sigmoid's: at this 300 Hz carrier the population is on the fast side of its crossover, and the generated source is the v_E^2 mixing of the carrier-frequency voltage excursion, operating-point independent to about 1% (§2.3, Fig. 7).

4.6 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input p, but p also shifts the operating point, entangling detection (σ'') with amplification. The sharper, mechanistic statement is that the *network coupling* sets the gain on its own. In the JR column To isolate amplification, we vary the JR global connectivity C —the (coupling strength J—and, for each C,) and solve analytically for the drive $p(C)$ that *pins* the operating point v^*, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\text{open}} = \frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ are held exactly constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows $\sim 30\times$ and the envelope resonance sharpens (Fig. 9a,b)—entirely , a network effect , since detection is frozen and only γ changes. The recovered gain $A_\Omega / A_{\text{open}}$ tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input p rather than the coupling C yields the same gain–γ relation, so the amplification depends only on the distance to criticality, not on which parameter delivers it. All these sweeps stay on the stable-focus side ($\gamma > 0$), approaching the supercritical Hopf from below: the $1/\gamma$ growth is the forced-resonance susceptibility of a damped network, not entrainment of an autonomous cycle. On the other side of the bifurcation the column oscillates on its own, and the demodulated drive instead phase-locks that limit cycle within an Arnold tongue (Fig. S4; also the entrainment side of Fig. 4 and the LaNMM and NMM2 maps)—: 1:1 frequency capture over a band that widens with drive, quantified by the locking range rather than a diverging $1/\gamma$ gain, the

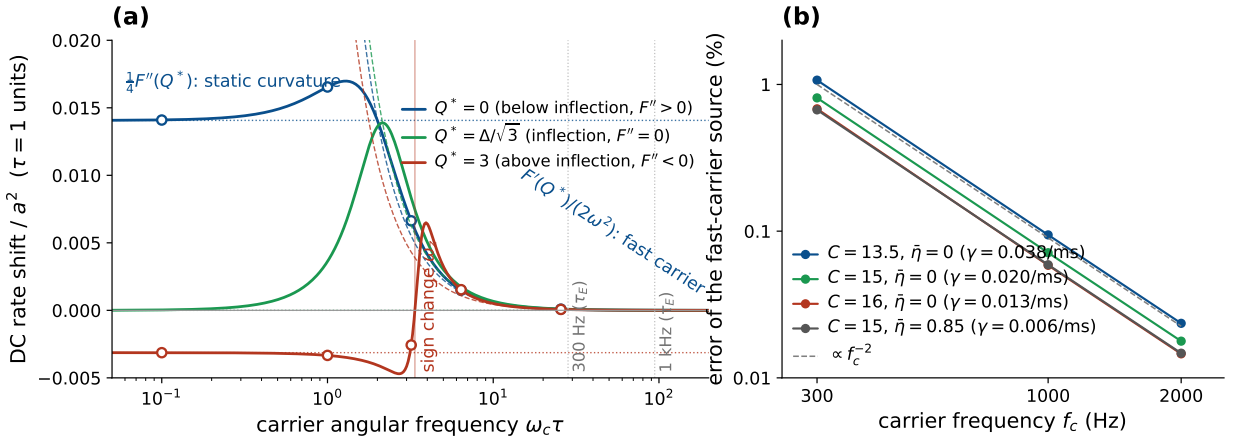


Figure 7: Static curvature at slow carriers, slope at fast ones. *(a)* Second-order DC rate shift of an uncoupled MPR population ($\Delta = 1$, $\tau = 1$) under a current carrier, Eq. (13), for operating points below, at and above the inflection of $F(Q)$. Dotted: the static limit $\frac{1}{4}F''$; dashed: the fast-carrier limit $F'/(2\omega^2)$; circles: RK4 integrations at $a = 0.05$ averaged over whole carrier periods (relative error below 10^{-3}). Above the inflection the sign changes at ω_z. The top axis converts $\omega\tau$ to a carrier frequency for $\tau = \tau_E = 15$ ms. *(b)* PING of Eq. (27): relative error of the fast-carrier difference-frequency source against the exact second-order response at four stable operating points (γ the damping of the gamma mode), two tones centered at 300, 1000 and 2000 Hz.

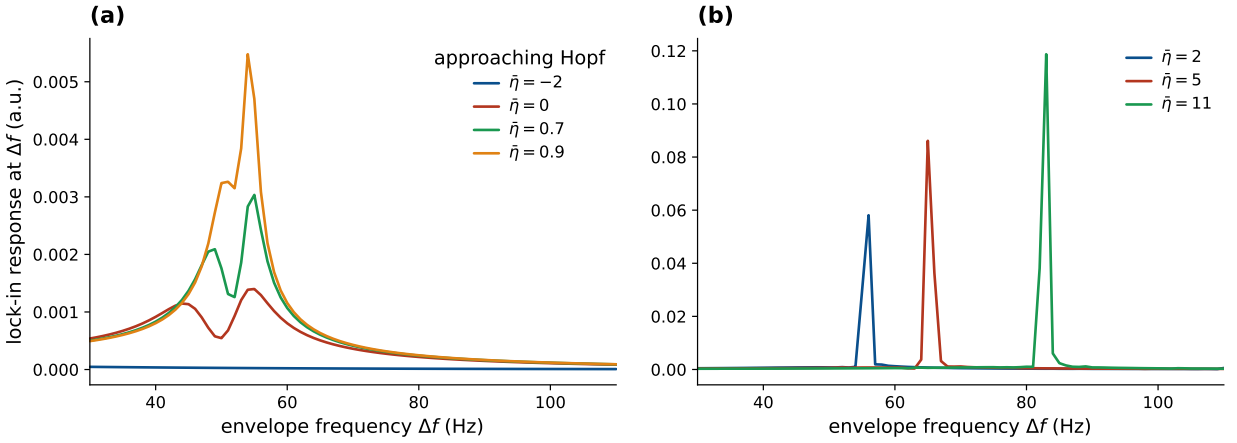


Figure 8: NMM2 PING gamma resonance (exact mean field, genuine two tones). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_H = 1.110$) forced resonance at the gamma frequency ($f_0 \approx 54$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_H$) entrainment of the autonomous gamma, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity, with no sigmoid anywhere in the model. Matched AM and two-tone drives agree at Ω to 0.07% well below the Hopf and to 4% at $\bar{\eta} = 0.9$, consistent with the JR surrogate control (Fig. 6d).

two coinciding only at onset. Because the drive is weak, on the autonomous side it biases phase (timing) more than amplitude, consistent with the timing-not-rate signature (§4.7).

The next-generation theory gives the same result without pinning. In the NMM2 PING the coupling C is the *literal* mean inter-neuron (QIF) synaptic weight J, and at this carrier the generated source is operating-point independent to about 1% (§2.3), so detection needs no pinning. Sweeping J at fixed background drive $\bar{\eta}$ toward the gamma Hopf J^*, the demodulated peak of r_E grows as $1/\gamma$ and saturates at criticality (Fig. S10): the heuristic JR result, reproduced with the coupling *and* the rectifier both derived rather than assumed, and a direct counterpart of the coupling-controlled weak-field sensitivity of QIF networks [24]. The two columns realize one band-

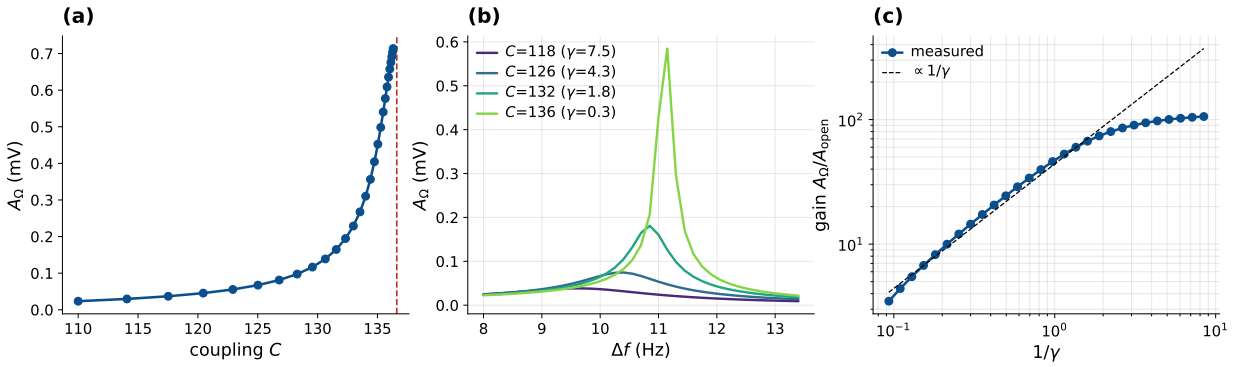


Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit). The operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling $C\,(= J)$ approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain A_Ω/A_{open} follows $1/\gamma$ (dashed) and saturates near the bifurcation.

agnostic law exhibit the same gain scaling in their two native bands—alpha bands, alpha (JR) and gamma (NMM2): the single-cell nonlinearity detects the envelope (coarse-grained into σ'', or the exact v^2), while the sharp, tunable gain is the network's—a resonant gain is a recurrent amplification set by proximity to the Hopf and adjustable through the coupling.

4.7 The resonance amplifies the envelope-locked modulation more than the mean rate

The demodulator feeds the network a constant term and a line at the beat Ω (Eq. (9)), and the transfer $G(\Omega)$ of Eq. (20) treats them differently: $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $G(0) \propto 1/\omega_0^2$ stays finite. Pinning v^* to hold the local detector curvature fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, we read both quantities from the same observable, the pyramidal rate $r_P = \sigma(v + s)$ with the field in its argument. The beat-locked modulation of r_P at $\Delta f = f_0$ grows 13.5× (26 to 346 mHz), tracking $1/\gamma$ before saturating near criticality; the shift of its mean grows too, 5.8× (−1.3 to −7.7 mHz), through rectification of the growing oscillation, but stays below 0.2% of the 3.9 Hz baseline (Fig. S11). Modulation exceeds mean shift by 19× far from the Hopf and by 45–66× near it. The network amplifies the envelope-locked modulation far more than the mean rate, so a small mean-rate change is the expected signature even where TI is acting, the mesoscopic counterpart of the observation that TI alters spike timing more than firing rate in the primate brain [8]. Finite is not zero: the mean does move, and the comparison must use commensurate measures, as here.

4.8 Microscale confirmation: TI realigns spike timing in the QIF network

The mesoscale prediction has a direct microscale test, since NMM2 is the *exact* mean field of an all-to-all network of quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities [22, 15, 24]. Integrating that spiking network ($N = 8000$ neurons per population, parameters identical to the PING mean field; Methods) shows TI bunching spike *times* into the beat cycle with a smaller change in the mean rate, on both sides of the gamma Hopf (Fig. 10). Below the Hopf (forced) the field-free network fires asynchronously at a flat rate; with TI on, spikes condense into envelope-locked bands (Fig. 10a,b). Above the Hopf (oscillatory) the free-running gamma is *gated* by the TI beat—forced beat, a forced cross-frequency modulation (phase–amplitude coupling, PAC), not frequency entrainment, since the ≈11 Hz detuning lies outside the 1:1 tongue: its bursts group into the $\Delta f = 42$ Hz envelope while the intrinsic ∼53 Hz gamma persists (Fig. 10c,d). Because the two rhythms coexist in a nonlinear medium, the QIF nonlinearity mixes them into intermodulation sidebands in the population rate—most prominently the direct Δf line and the sum component near 95 Hz (53+42), with the difference tone near 11 Hz (the detuning itself);

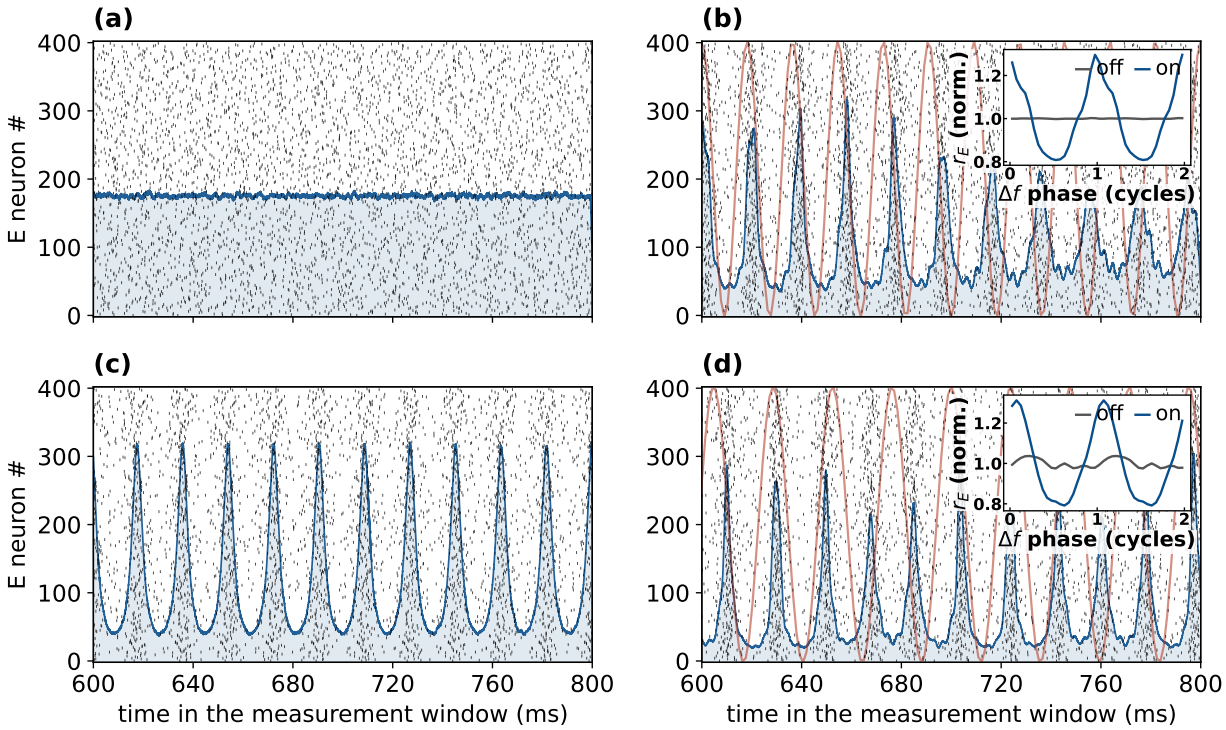


Figure 10: QIF spiking network: TI bunches spikes into the beat cycle with a smaller change in mean rate. $N = 8000$ excitatory and inhibitory quadratic integrate-and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (27)). The circuit's intrinsic rhythm is gamma (∼53 Hz), so the beat is set in the gamma range (carrier $f_c = 300$ Hz, genuine two tones). Black ticks are E spikes (400 of 8000), blue the population rate, red the TI envelope; the bifurcation parameter is the common drive $\bar{\eta}$ (gamma Hopf at ≈ 1.1). *(a,b)* Forced ($\bar{\eta} = 0.85$), TI off then on at $\Delta f = 55$ Hz: asynchronous firing becomes envelope-locked bunching. *(c,d)* Oscillatory ($\bar{\eta} = 1.5$), TI off then on with the beat detuned to 42 Hz: the free-running gamma persists but gains a Δf-locked component. Because the ≈11 Hz detuning places the beat outside the 1:1 tongue, this is forced cross-frequency *gating*, not frequency capture (contrast Fig. S4); the detuning is what certifies the locked component as a driven effect. The last 200 ms of the 800 ms measurement window are shown. *Insets* in *(b,d)*: the population rate folded on the beat phase over the whole window (44 and 34 cycles), each normalized to its own mean (off gray, on blue); with that many cycles the near-commensurate 53 Hz gamma averages out of the 42 Hz fold in *(d)*. Both are flat with TI off.

this is the spiking-network fingerprint of the same nonlinear frequency mixing that operates at the macroscale, and it is the sideband signature of *gating* rather than the frequency pulling of entrainment. The spiking network reproduces the exact NMM2 mean field's gamma spectrum (Fig. S12a). Across both regimes the mean rate rises by 8–10%, while the Δf modulation depth of the population rate, $A_\Omega/\langle r_E \rangle$, rises from below 0.003 to 0.29 (forced) and 0.23 (gated) (Fig. S12c), measured over 44 and 34 whole beat cycles after 800 ms of settling (Methods). Timing modulation therefore exceeds the relative mean-rate change by a factor of about three at the population level. Per neuron the locking is weak: the pairwise phase consistency of single-neuron spike phases at Δf rises from its field-free control (median −0.02) to a median of 0.003–0.005, and the pooled vector strength from below 0.02 to 0.12–0.13. The population's beat-locked component is built from many weakly biased cells, not from strongly locked ones. This is the spiking-level counterpart of §4.7 and a qualitative analog of the in-vivo finding that TI alters spike timing more than firing rate [8].

4.9 Falsifiable predictions and tests

The mechanism makes four predictions, each with an experimental handle.

1. **Carrier** independence at fixed polarizationdependence separates access from mixing**.** Once the post-

coupling polarization $\varepsilon = \lambda_f E_0$ is matchedIn the static limit and in the fast-carrier QIF asymptote, the leading quadratic beat component no longer depends on beat is independent of f_c at fixed local polarization, provided the carrier lies outside the responsive bands of both detector and network ; frequency dependence reappears through the effective coupling $\lambda_f = \lambda_0 |H_c(f_c)|$, or when a fast cellular or network stage is itself resonant (§E). The *applied*-field threshold still rises with f_c through the membrane roll-off does not excite another network mode. At fixed applied field, access restores frequency dependence (Eq. (22))—as single-neuron accounts also predict [5, 2]. *Test:* hold the in-zone polarization fixed (raising E_0 to compensate the measured λ_f) while sweeping f_c; the network signature is a flat envelope response. At fixed applied field the same sweep reads the entry point: the roll-off should sit in the kHz range (AIS access, Na_v gating), not at the ∼10 Hz somatic corner (§2.6). Existing AM-versus-tACS comparisons match the extracellular modulation rather than ε [8], so this is the correction they lack; detector dynamics can add it near the crossover. *Test:* measure or estimate local polarization during a carrier sweep and compare the response before and after correcting for access. The remaining frequency dependence constrains the mixing mechanism. Its cutoff is to be measured, not assigned to a particular compartment in advance.

2. **State- and carrier-dependent response.** The generated beat depends on the operating point through the detector coefficient, and the two modeled detectors disagree on how: the memoryless sigmoid follows $\sigma''(v^*)$ and reverses sign across the inflection, where PEIX (Eq. (24)) peaks; the dynamic mean field at a fast carrier follows F' and the network susceptibility, without a sign rule (§2.3). *Test:* shift v^* with tonic drive , neuromodulation, or task state at a fixed beat, and sweep the carrier at a fixed beat vary tonic drive and carrier frequency while measuring both amplitude and phase of the beat response. Compare these with the predicted mixing coefficient after accounting for changes in network susceptibility; a response-amplitude change alone does not isolate the detector. A sign reversal across the operating point would select the memoryless description, a monotonic change the dynamic one.

3. **Transduction class.** A weak smooth nonlinearity transduces A^2 and an envelope follower transduces A (§2.4), and the two are separable at a single matched target. The former gives $Y_\Omega \propto \varepsilon^2$ with a second-harmonic ratio vanishing as $\varepsilon \to 0$; the latter gives $Y_\Omega \propto \varepsilon$ with the ratio pinned at 1/5. *Test:* sweep field amplitude off resonance (so that both lines sit on the flat part of G) and read the log–log slope together with $Y_{2\Omega}/Y_\Omega$; on resonance the network suppresses the second harmonic by $\sim 3\mathcal{Q}$ and the test must be corrected by an independently measured $G(2\Omega)/G(\Omega)$.

4. **Frequency selectivity and state-dependence.** The response is largest when Δf matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation, so manipulations that move the operating point or the distance to the Hopf—psychedelics, PV-interneuron dysfunction, attention, arousal—should scale, and can abolish or sign-reverse, TI efficacy . Changes in operating state can therefore alter both the generated drive and the recurrent gain [13]. *Test:* sweep Δf through a region's intrinsic band across brain states and track the resonant peak and its width.

Table 3 confronts these predictions—together with two corollaries the mechanism already shares with published observations—against compares the predictions and related model consequences with existing evidence. Two are qualitatively consistent with published observations: TI alters spike timing more than firing rate in vivo [8] (our $G(\omega_0) \gg G(0)$ asymmetry, §4.7–4.8), and most pyramidal TI responses vanish when synaptic transmission is blocked [9] (a network read-out, which tests network involvement rather than excluding a surviving cellular detector). A quantitative reanalysis of these datasets against the model is a natural next step, and §5 specifies the amplitude-exponent fit it would require.

5 Discussion

The models show how neural populations can act as tuned receivers of TI inputs. Nonlinear mixing generates a component at the difference frequency , and recurrent dynamics selectively amplify it near a network resonance. Generation and amplification are distinct contributions: a large network response does not, by itself, imply efficient detection. This

Prediction (model basis)	**Existing evidence**	**Sharp test**
Transduction class: $Y_\Omega \propto \varepsilon^2$ and vanishing $Y_{2\Omega}/Y_\Omega$ (square law) vs. $\propto \varepsilon$ and 1/5 (envelope following)	— (untested; existing comparisons match extracellular field, not ε)	Amplitude sweep off resonance; transfer-corrected harmonic ratio
Timing-dominated response: the envelope-locked modulation of the rate grows toward the Hopf faster than its mean ($G(\omega_0) \gg G(0)$)	*Qualitatively consistent*: TI alters spike timing more than firing rate, primate [8]	Lock-in of spike timing at Δf vs. the mean-rate change, in the same effect measure
Network read-out (population nonlinearity + synaptic recurrence)	*Qualitatively consistent*: most pyramidal TI responses vanish under synaptic block [9]	Synaptic block should abolish the demodulated response reduce recurrent amplification; cellular mixing may persist
Carrier independence at fixed polarization (no ω_c in A_Ω)	TI thresholds inherit the carrier's f_c-dependence [5, 6]; bare kHz weakly effective [36]	Fix in-zone polarization, sweep $f_c \to$ flat envelope response ; at fixed field, a kHz-range (not 10 Hz) roll-off in the stated limits; fit access and residual detector dependence
State- and carrier-dependent response (the detector coefficient is model-dependent)	Cell-type-specific TI responses (PV vs. pyramidal) [9]	Shift v^* by state at fixed beat; sweep f_c at fixed beat
Frequency selectivity, sharpened $\sim 1/\gamma$ near criticality	— (a sharp new prediction; largely untested) Quantitative gain–bandwidth relation not tested here in biological data	Sweep Δf through a region's band across brain states

Table 3: The mechanism's predictions confronted with existing in-vivo / in-vitro evidence. Entries cite documented qualitative findings; quantitative reanalysis is left to future work.

complements rather than competes with the single-neuron The population models separate the generation of a difference-frequency drive from its recurrent amplification. Cellular mixing can supply that drive, while the network determines its frequency selectivity and gain. This interpretation connects the ion-channel account [2]: both are rectify-then-filter demodulation, ours is agnostic about which fast cellular nonlinearity supplies the mixing, and it inherits the brain's own resonances, so a region's tuning is set by its intrinsic dynamics as a radio is tuned by its tankmechanism [2, 4] to the network dependence seen in slices and simulations [9, 10, 11]. It does not require cellular mixing and circuit amplification to be competing explanations.

The detector's dynamics matter. The The distinction between a static sigmoid and the dynamic a dynamic population functional is a time-scale distinction. In the slow-synapse limit, the exact QIF mean field both support quadratic mixing in the weak-response regime, but their mixing coefficients and operating-point dependence differ (§2.3). A steady-state firing-rate curve therefore does not, on its own, determine the response to a fast carrier. The common mechanism is nonlinear mixing followed by network tuning, not a universal sigmoid-curvature law. The single cell need only supply a fast nonlinearity that samples the carrier; in the exact mean field that is the spike-generating quadratic acting on a voltage excursion of order a/ω. A population transferfunctional calibrated to the relevant cellular dynamics could incorporate ion-channel mixing. A static firing-rate sigmoid specifies only the steady-state relation; its curvature reduces to the nonlinear curve in Eq. (5) while synapses retain their dynamics. If relaxation is also fast relative to the carrier, the curve's curvature gives the leading beat. Otherwise, the dynamic mixing calculation is required. The curve's slope has a separate role: it converts a slow input into a rate change and sets the local gain entering recurrent feedback. This reduction explains what can be included in a population transfer, but does not establish that dynamic functional or its carrier-frequency dependence.a particular QIF

parameterization reproduces the channel mechanisms measured by Luff et al. [4]. That identification requires matching the relevant dynamical response, not only a steady-state firing-rate curve.

Where the carrier-sampling nonlinearity may live. The literature points to three non-exclusive loci. (i) *Axonal Na^+ channels* at nodes Fast access has several possible anatomical and electrical origins. Nodes and the AIS , where fast contain voltage-gated Na^+ rectifies because activation outpaces inactivation [2, 37]and inactivation-mediated block passes the slow envelope while removing the carrier [5]. (ii) *Presynaptic terminals*, which polarize ∼2–10× more channels capable of nonlinear rectification [2, 37]. Presynaptic terminals can polarize more strongly than somata [38, 39, 40]and convert depolarization into release through a steeply nonlinear Ca^{2+} relation. Sensitivity is not speed, however, but small size alone does not imply fast charging: for an isopotential compartment $\tau_m = R_m C_m$ with both quantities per unit area, so the time constant does not shrink with size—a small bouton has small capacitance and correspondingly large input resistance . A terminal therefore tracks a kHz carrier no better than a soma, and the short axonal time constants come from high-conductance nodal membrane rather than from small size. Terminals are the *sensitive* locus, not the fast onemembrane, the product of specific resistance and capacitance is independent of area. Conductance, geometry, and the carrier sampling is supplied by (i). (iii) *Threshold spiking*, where a weak bias near threshold is rectified through stochastic resonance [30, 29]. Each is a candidate; the simulations impose an access filter and a nonlinearity, they do not identify the compartmentmode of stimulation determine the response.

The periaxonal space provides another candidate access route. Cohen et al. [41] combined recordings, electron microscopy, and cable modeling to establish a conducting space between the axon membrane and myelin, established a conducting periaxonal space approximately 12 nm wideand incompletely , partly sealed at the paranodes. Their double-cable model separates the potentials across the myelin sheath and the underlying axon membrane. This architecture motivates computing the transfer from an applied extracellular kHz field to the local voltage at a candidate mixer. The saltatory-conduction measurements do not determine that transfer. , and modeled the distinct potentials across myelin and the axonal membrane. Their methods report a minimum fitted axonal decay time near 1 ms; this is not a measured TI access time. Luff et al.'s PWM-TI model likewise includes a 1 ms passive response within a 1–100 ms parameter sweep [42]. Separately, Semenov et al. [7] measured field-induced membrane charging near 0.7 μs in isolated neurons under strong, microsecond field pulses. This is a different response mode from intracellular current charging or periaxonal conduction. Together these studies motivate examining a shorter effective access time, while leaving its value and field coupling preparation-dependent. A 10–100-fold shorter effective carrier-access time scale reduction in a QIF reduction remains a modeling hypothesisto test against this geometry. Changing the QIF membrane time scale also requires recomputing its operating point and resonancemembrane time is therefore a sensitivity hypothesis. Its population-level consequence can be studied through the existing slow-synapse limit, with each physical filtering process represented once. filter included once; a new periaxonal circuit is not needed for that limiting argument.

The conventional TI dose map is an assumption about physiology, not a consequence of field physics. The envelope is a mathematical property of a high-frequency waveform. It is not itself a low-frequency field, and nothing in Maxwell's equations delivers it to a neuron; biology must construct it. Targeting work since [1] takes the A spatial TI dose map depends on this physiology. The usual envelope-modulation amplitude $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$ as the spatial dose, which assumes $R(\mathbf{x}) = \Phi(\mathcal{M}[A])$ for monotonic Φ—that the response depends on measure $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$ assumes that neural response is determined by the envelope excursion*alone*, and not separately on the carrier baseline, the operating point or the orientation. That assumption is substantive, and a weak smooth nonlinearity cannot satisfy it: . In the local square-law regime, the beat instead depends on $\varepsilon_1\varepsilon_2$, and the carrier average Eq. (16) is invariant under $A \to -A$, so its expansion contains only even powers and no also contains a DC term proportional to A exists. Square-law transduction is therefore the generic perturbative answer and envelope following is the special case needing justification, since recovering A requires rectification (§2.4) and any *smoothed* threshold reverts to A^2 as the field weakens. $\varepsilon_1^2 + \varepsilon_2^2$. Orientation, access, operating point, and network susceptibility further shape the response. For a smooth memoryless transfer, carrier averaging is even in the envelope amplitude, giving quadratic transduction when the second-order coefficient is nonzero. Dynamic nonlinearities can also mix at second order, with frequency-dependent coefficients. A sharp rectifier can instead approximate envelope following over a finite amplitude range; smoothing or heterogeneity can restore quadratic behavior at smaller inputs. The amplitude and harmonic tests in §2.4 distinguish these regimes.

Different field strengths and operating states may recruit different nonlinear mechanisms; the present analysis addresses the weak-response regime. Near threshold a sharp channel or spike nonlinearity may act as a rectifier; well below it, the leading responseof any smooth nonlinearity, memoryless or dynamic, is quadratic mixing, filtered and amplified by the network as described here. Which of the two operates at human field strengths is an empirical question that present evidence does not settle. Primate recordings at human-comparable fields find an AM waveform less effective than matched-depth tACS, but do not estimate the difference closely enough to select a transduction law [8], and peripheral-nerve work argues that some TI responses reflect carrier integration rather than envelope extraction [6]. Two 2025 studies sharpen that point at suprathreshold intensities: in peripheral nerves the strength–frequency curves of TI, amplitude-modulated Stimulation intensity can also change the mechanism. At high intensities, TI and unmodulated kHz waveforms overlap [5], and in isolated hippocampal neurons pure 2 and 20 kHz fields excite at high intensity while modulation patterns the firing without fields can have similar excitation thresholds, with modulation patterning firing rather than lowering the threshold [7]. At those intensities the carrier itself stimulates. The weak-field mixing route analyzed here is a candidate for the subthreshold, modulatory regime of the primate recordings and of the recent patient studies of hippocampal TI [8, 43]. What settles it is not whether TI works but which quantity it transduces, and §2.4 gives the measurement. [5, 7]; some peripheral responses do not require envelope extraction [6]. The weak-input calculation here concerns subthreshold mixing and its amplification. It provides a candidate mechanism for modulatory responses, not a common explanation for every TI preparation [8, 43].

The amplifier is not specific to TI. A tACS field oscillates in the envelope band directly and needs no demodulation, but it drives the same near-Hopf resonator. Sweeping the coupling toward the alpha Hopf at *pinned* operating point , a direct Δf drive reproduces The recurrent amplifier is shared with direct low-frequency stimulation. In our coupling sweep, holding the JR operating point fixed holds the local curvature fixed while the network gain changes. A direct in-band drive shows the same $1/\gamma$ growth and resonance sharpening as the demodulated TI drive, linear in the field rather than square-law (Fig. S3). Coupling-controlled gain is known: the dynamical transfer function of an E–I network is known to become resonant near a Hopf bifurcation, with a peak that *diverges* on the bifurcation line as connectivity is varied [44], and increased self-connectivity is known to enhance weak-field resonant sensitivity in the exact mean field [24]. What the pinned sweep adds is the separation of amplification from *detection*. Holding the local detector curvature $\sigma''(v^*)$ constant while the network transfer changes isolates the network gain, and shows that the same amplifier acts on a drive the population has itself generated. TI then differs from tACS by a generating front end that converts a kHz carrier pair into an effective in-band drive; downstream, both share the criticality- and coupling-controlled amplifier, so, matched for effective in-band drive, read through the same observable and near the same state, they should entrain alike. The cancellation is conditional on those matches.

Amplification is critical slowing, of a mean-field kind. Gain $\propto 1/\gamma$ as the damping vanishes at the Hopf is the diverging susceptibility of a system approaching a bifurcation. In the exact next-generation mean field the reduction from the spiking population is exact in the thermodynamic limit and (r, v) are order parameters [45, 22], so the Hopfis a macroscopic transition; because the reduction assumes all-to-all coupling the critical point is mean-field, with , consistent with the known resonant susceptibility of recurrent excitatory–inhibitory networks [44, 24]. Near the Hopf, this is temporal critical slowing but no in a mean-field model, not evidence of a diverging spatial correlation length. The coupling sweep moves the model toward the bifurcation; the stimulus probes its response.

Prior work already places the operative nonlinearity in the network. Recurrent excitation, with short-term synaptic depression, accounts for tACS entrainment in a large-scale spiking model and yields an Arnold tongue over stimulation amplitude and frequency [46]; the dynamical transfer function of an E–I network, in both rate and spiking form, is resonant near a Hopf bifurcation with a peak that diverges on the bifurcation line as the connectivity is swept [44]; an amplitude-modulated high-frequency drive phase-locks a cortical model to its envelope through the same target engagement as low-frequency tACS [47]; and comparing neuron models directly, integrate-and-fire membranes fail to reproduce observed TI behavior while Hodgkin–Huxley ones succeed [48]. The general case for stimulation as interaction with ongoing dynamics has been made explicitly [49]. What we add is the separation of the generated source from the network response , the comparison of a memoryless and a dynamic detector, Related models already reproduce tACS entrainment and , downstream of both, that the transduced quantity in the weak-response regime is A^2 rather than A, a distinction the cited network studies do not draw. AM-driven phase locking [46, 47, 49]. The separation here is between the drive generated by mixing and the response to that drive. When TI and tACS are matched for effective in-band input, observable, and operating state, they share the amplifier. Resonance alone therefore gives TI no

efficiency advantage.

Estimated response magnitude and comparison with tACS. Two obstacles separate a transcranial field from a response at Δf, and they need different answers. The first is that weak fields barely move a membrane: measured coupling is Absolute response estimates remain sensitive to access and transduction. Measured static field-to-voltage couplings are 0.12–0.21 mV per V/m in hippocampus [50, 51] and up to ∼ 0.5reach about 0.5 mV per V/m for in cortical L5/6 pyramidal cells [52], human cortex sees about ; human cortical fields near 0.8 V/m at 2have been reported at 2 mA [53, 54], and most applied current is shunted through scalp and skull [55]. The field shifts a soma by a few tenths of a millivolt against the ∼ 23 mV separating rest from threshold [56]. This is as true of tACS as of TI. The second belongs to TI alone: its field carries no power at Δf, so the response there must be generated by a nonlinearity. Resonance addresses the first, a population near its own bifurcation responding to a weak push far more strongly than an isolated cell, but not the second, because it amplifies TI and tACS alike and cancels when the two are compared under the matching conditions above. These static couplings do not fix the coupling at kHz. For equal TI carrier polarizations ε, the JR beat drive is $\sigma''\varepsilon^2/2$, whereas a direct drive of the same peak polarization gives $2\sigma'\varepsilon$. Their leading amplitude ratio is $\eta_{\text{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$: the relative efficiency decreases with input amplitude in this perturbative regime. A larger recurrent gain amplifies both drives.

The resonance contributes a factor of order ten. With $\tau = 1/\gamma$ the reverberation time , the gain at resonance relative to the same circuit's response to a steady input is $\mathcal{Q} = \pi f_0 \tau$. Cortical population activity relaxes For scale, a reverberation time $\tau_{\text{rev}} = 1/\gamma$ gives a near-Hopf quality factor $\mathcal{Q} \simeq \pi f_0 \tau_{\text{rev}}$ and power bandwidth approximately $1/(\pi\tau_{\text{rev}})$. Population relaxation over a few hundred milliseconds [57] , giving $\mathcal{Q}$ of order 10 at $f_0 = 10$ Hz—an order of magnitude, not a value, since that timescale is wideband rather than alpha-specific. Gain also costs bandwidth : the resonance is $1/(\pi\tau)$ wide in power (full width at half maximum), about would give a quality factor of order ten at 10 Hz and a bandwidth of order 1 Hz, so a Δf fixed at a group-mean alpha may sit outside it; whether an individual's alpha drifts that far within a session is not settled by this estimate. At equal field strength TI is far weaker than tACS for the memoryless detector at matched peak polarization. With per-carrier polarizations $\varepsilon_{1,2}$ the demodulated drive is $\frac{1}{2}\sigma''\varepsilon_1\varepsilon_2$ against $2\sigma'\varepsilon$ for tACS at the same peak polarization, a ratio $\eta_{\text{det}} = \sigma''\varepsilon/4\sigma'$ proportional to ε: a square-law detector gets *less* efficient as the signal weakens. Formally η_{det} reaches unity near $\varepsilon \approx 7$ mV, but that is the order of the distance to threshold and lies outside the expansion's validity, so parity is not reachable by this route rather than merely expensive. At 1 V/m with a 1 kHz carrier—now comparing at matched *applied* field, so the carrier's membrane attenuation enters twice while the in-band tACS drive is unfiltered—TI comes out ∼ 300× weaker than tACS if the carrier reaches a fast nodal compartment, and ∼ 10^6× weaker if it reaches only the soma. . This is an order-of-magnitude comparison: the measured relaxation is not specific to the alpha mode. Narrow amplification also makes detuning consequential.

The one in-regime measurement sits well above this estimate. Primate measurements illustrate both the relevance and the limits of an efficiency comparison. Vieira et al. worked at [8] used modulation depths of 0.5–0.62 V/m , chosen deliberately to match human studies rather than the 60–383 V/m of the rodent literature, and report TI and reported TI effects at 0.34 times the strength those of conventional tACS (95% CI 95% confidence interval 0.04–0.65), or ∼ 12% per unit current under their stated assumption that effects are roughly proportional to applied current [8]. That normalization is not neutral here, since a square-law drive scales as I^2 while tACS scales as I; the per-current figure is therefore transduction-class-dependent and we do not use it. Their AM-versus-tACS comparison is the most direct available. Having measured that transcranial current is shunted increasingly with frequency, they compensate for it and deliver conventional tACS and an AM waveform through the same electrodes, with the . Their within-electrode AM comparison compensated for frequency-dependent current shunting and matched extracellular modulation depth near the recorded cellscalibrated to 1.14–1.17 times the tACS amplitude. Against a baseline PLV of 0.043, tACS raises the median to 0.20 (95% CI 0.043–0.64) and AM-tACS to 0.10 (0.023–0.38), a paired difference significant at $p = 0.01$.

Taking their fields with a favorable cortical coupling (∼ 0.5 mV per V/m) through a fast element at 2 kHz gives $\varepsilon \approx 0.1$ mV and hence $\eta \approx 0.8\%$, far below the ordering they report; a somatic route is worse by two further orders. Three limits keep this from being a quantitative test. The comparison rests on A favorable fast-access estimate with $\varepsilon \approx 0.1$ mV gives only about 0.8% for the JR detector, below the reported ratio. This discrepancy is not a calibrated rejection of that detector: the comparison involved twenty neurons, of which ten responded with ten responding to at least one modality and seven to both, so the sample is selected for responsiveness and the reported intervals

are correspondingly broad. PLVis a downstream statistic that ; phase-locking value (PLV) depends on baseline locking , network susceptibility and state as well as on the generated drive, and in their full dataset baseline PLV predicts the sign as well as the size of the change. And the quantity matched was the extracellular modulation, not and network state; and the post-coupling polarization $\varepsilon = \lambda_f E$ of §2.2 at whichever compartment supplies the nonlinearity, so the ratio reports the whole end-to-end path from applied field through frequency-dependent access, transduction, network dynamics and spike-phase statistics. It does not isolate the transducer. What the experiment establishes is that a high-frequency AM waveform drives phase locking less effectively than a matched extracellular fields do not ensure matched polarization at the cellular mixer. An unspecified nonlinear relation between generated drive and PLV also prevents identification of the input-amplitude exponent. The data establish a weaker AM effect than the matched-depth low-frequency one, and that the prominent population effect was often *reduced* locking to ongoing activity rather than increased beat locking, a phenotype richer than the increased envelope-locked modulation our models show; the magnitude of that difference is not estimated precisely enough to discriminate among candidate transduction lawsdrive, but do not isolate the transduction law.

For a memoryless transfer the ratio is bounded whatever the operating point: by $\rho\varepsilon/4$ at fixed slope, and by 0.348 over the whole logistic family , a ceiling approached only where the transfer has narrowed to a microvolt width comparison, the matched-peak efficiency over the logistic family has a supremum of approximately 0.348 (App. D). Brain state moves the efficiency but cannot lift it off that ceiling.

A sigmoid steep enough to approach the ceiling is a step detector , which is neither the square law nor the linear envelope follower of §2.4: its harmonic ratio at the optimum is Approaching it at weak fields requires a transfer curve narrowed to microvolt scales; the limiting detector is step-like and has a second-harmonic ratio near 0.15, not 1/5 (App. D). Which transduction operates in vivo is not settled by the evidence available. The ceiling is a supremum over a family of memoryless transfers at matched peak polarization, not a prediction for any one preparation, and it does not govern the dynamic detector of §2.3; its numerical proximity to a reported ratio is neither support nor exclusion. Vieira et al. read their own data as favoring rectification, invoking the radio analogy explicitly and proposing asymmetry between inward Na^+ and outward K^+ channels as the operative nonlinearity. They also note that circuit-level frequency dependence could contribute to demodulation, which is the mechanism developed here.

A retrospective test on existing primate data would need the link between generated drive and the phase-locking value (PLV) fixed or calibrated: with an unspecified monotone link the amplitude exponent is not identifiable, since the link absorbs it. Prospectively, matching post-coupling polarization rather than extracellular field alone, and reading the amplitude slope together with the transfer-corrected harmonic ratio of §2.4, separates the effective response laws without an absolute calibration.

The amplification half is separate, for a definite reason. The J-curve reports A_Ω/A_{open}, the closed-loop response normalized by the open-loop detector output at the same operating point. The detector divides out by construction: whatever manufactures the beat component—a smooth population curvature, a nodal rectifier, a spike threshold—the recurrent network's response to that in-band drive is the same, and it is that response which follows $1/\gamma$ and sharpens toward the bifurcation. The same holds in the exact mean field, where the rectifier is the derived v_E^2 term and no operating-point pinning is needed at all. Detection and amplification are therefore separable in experiment as well as in principle, and today's evidence bears only on the first. The population contribution established here is the tuning and amplification ; detection may occur microscopically or through the effective population transfer.

Whether distinct from both weak square-law transduction operates in the human brain at safe field strengths is open. What the parity argument establishes is narrower: any smooth weak nonlinearity with a nonzero second-order response transduces A^2, so a square-law route exists and is the generic leading perturbative form, not a quantitative estimate of the in-vivo response. The evidence available is consistent with a more rectifying transduction in vivo, in which case the detector is cellular; it does not establish that . The amplification half is unaffected either way: whatever generates the beat component, a network near its bifurcation amplifies it as $1/\gamma$. The amplitude sweep of §2.4 distinguishes these effective response laws under the stated controls: a slope of 1 with a harmonic ladder near 1/5 indicates envelope following, a slope of 2 with a vanishing ladder the square law, and a step-like threshold gives intermediate ratios (App. D). detection and linear envelope following. This bound applies to that memoryless family and matching convention, not to the dynamic QIF detector or to all cellular mechanisms. Its proximity to an experimental ratio is not a fit.

Limitations. The field enters the population transfer directly; we treat the field-to-nonlinearity coupling λ_f phenomenologically rather than modeling channel kinetics, and ε should be read as post-coupling polarization. The population sigmoid is a coarse-grained stand-in for distributed The models leave identifiable limitations. Access is phenomenological, and the QIF quadratic does not replace channel kinetics; single-cell nonlinearities. JR alpha near this Hopf is high quality-factor, so the stable-side peaks are narrow and would be broadened by noise and population heterogeneity ; the columns here are single, deterministic and noiseless, and coupled noisy columns with an explicit access filter $\lambda_f(f_c)$ remain the natural next stepcomparisons find TI responses that Hodgkin–Huxley models reproduce and integrate-and-fire models do not [48]. Our spiking-network result establishes population mixing and amplification in QIF, without resolving that biophysical difference. Noise, heterogeneity beyond the assumed excitability distribution, and interactions between columns can broaden or shift the deterministic resonances. Cascading ordinary neural-mass transfers does not synaptic filters cannot solve carrier access, since the intervening synaptic filter removes the kHz component before a second sigmoid can rectify it. A population of microscopic rectifiers could supply the effective detector, but heterogeneity in their thresholds would broaden the aggregate transfer, and whether it stays sharp enough at human fields is a separate biophysical question. The exact mean field resolves the static-versus-dynamic question for a QIF population with an instantaneous quadratic; real spike initiation has gate kinetics of 0.1–0.2 ms, so at 1 kHz the gate is not a small parameter, and the fast-carrier coefficient of §2.3 is a model statement, not a measured cellular one; single-compartment modeling finds that channel gating kinetics, not the current–voltage nonlinearity , carry the TI response, and that integrate-and-fire cells fail to reproduce it[48]. That negative result concerns an isolated cell, whose slow stage must be intracellular. In a population the synaptic and recurrent dynamics supply the slow stage, and the cell need only rectify the carrier: a network of quadratic integrate-and-fire cells responds to TI (§4.8) with the spike-initiation quadratic as the rectifier, and a population of hard-threshold cells with spread thresholds would respond through the fraction crossing threshold at the carrier peaks, linearly in the envelope above the spread and quadratically below it. The human evidence is also not uniformly positive: a controlled study found peripheral but no because they attenuate the carrier before a subsequent nonlinearity receives it. Human results are also mixed, including reported peripheral effects without central effects [58] , and a systematic review reports that and null occipito-parietal TI did not modulate alpha power [59]. The narrowband prediction turns the second into a design question—whether Δf tracked individual alpha closely enough to stay inside a $\sim$ 1 Hz response band—which is a prediction to be tested, not an explanation to be assumedalpha-power modulation [59]. Testing the proposed mechanism requires measuring access, generated beat, and network tuning in the same preparation.

6 Conclusion

The models show how neural populations can act as tuned receivers of temporal-interference inputs. Nonlinear mixing generates a component at the difference frequency, and recurrent dynamics selectively amplify it near a network resonance. Generation and amplification are distinct contributions: a large network response does not, by itself, imply efficient detection.

The detector's dynamics matter. A memoryless firing-rate sigmoid and a dynamic quadratic-integrate-and-fire mean field both support quadratic mixing in the weak-response regime, but their mixing coefficients and operating-point dependence differ. A steady-state firing-rate curve therefore does not, on its own, determine the response to a fast carrier; the crossover from curvature to slope sits at the population's own frequency, below the carriers TI uses. The common mechanism is nonlinear mixing followed by network tuning, not a universal sigmoid-curvature lawstatic and dynamic population descriptions are connected by their time scales. Fast membrane relaxation gives the NMM2 slow-synapse limit: a nonlinear static rate map coupled through dynamic synapses. Its curvature describes carrier mixing when relaxation is also fast relative to the carrier. Outside that limit, the dynamic response must be used. Fast cellular access and slow recurrent resonance can therefore coexist, with each physical filtering process represented once.

This distinction changes how the stimulation should be interpreted. In the local In the weak square-law descriptionregime, beat generation depends on the product of the two effective carrier inputsrather than , not only on the envelope excursion, and network susceptibility then determines how strongly that generated component appears in the response. When . TI and a direct low-frequency drive excite the same effective input and observable around the same state, they share the downstream amplifier , when input, observable, and state are

matched, so resonance alone does not give TI an gives TI no efficiency advantage. Which compartment passes a kHz carrier to the mixing nonlinearity is a hypothesis in an access-filter model, and the absolute response magnitude The anatomical access pathway and absolute response at human field strengths remains to be established. The analysis identifies what should be measured separately: carrier access, nonlinear beat generationremain open. Measuring access, mixing, and circuit tuning separately tests the proposed mechanism without attributing all three to a single transfer curve.

Code and data availability

All simulation and figure-generation code, the committed datasets, and a one-command reproduce script (`run_all.py`) are openly available at the paper's GitHub repository, archived at Zenodo (doi:10.5281/zenodo.21009618). The repository `README` documents the per-figure script mapping (a figure→script table) and the dependency order, so every figure in this note can be regenerated from the committed inputs. The code is released under CC-BY-4.0.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

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A Notation and units

These neural-mass models are phenomenological: potentials are in mV, firing rates in s^{-1} (≡Hz), time constants in ms, and frequencies in Hz, unless noted; a "–" marks a dimensionless quantity. A few glyphs are reused across the four models (JR, LaNMM, NMM2/MPR, QIF) by long-standing convention; these are disambiguated by context and flagged below. In particular, A without an argument is the excitatory PSP *gain* in the JR/LaNMM synapses; $A(t)$ is the instantaneous envelope of the two-tone field and A_Ω a lock-in amplitude. The applied-field modulation amplitude is ε in JR and LaNMM alike, and A_f in NMM2/QIF. These are not interchangeable units. In JR and LaNMM the field is a membrane polarization in mV entering the sigmoid argument, so ε is measured against the sigmoid voltage scale $1/\rho \approx 1.8$ mV, which is the same in both models. In NMM2/QIF the field enters the excitatory membrane current in the dimensionless MPR units of v_E, and A_f carries no volt scale at all. Likewise C is the JR connectivity constant and the NMM2 global synaptic gain $C\,(\equiv J)$.

Symbol	Meaning	Units
Firing-rate nonlinearity (sigmoid)		
$\sigma(v)$	population firing-rate transfer function	s^{-1}
$\sigma'(v)$	sigmoid slope (read-out gain)	$s^{-1}\,mV^{-1}$
$\sigma''(v)$	sigmoid curvature; the memoryless detector coefficient	$s^{-1}\,mV^{-2}$
$F(Q)$, F', F''	MPR steady-state rate curve, Eq. (5), and its derivatives	–
$C(Q,\omega)$, $K(\omega)$	MPR second-order DC coefficient at fixed current, Eq. (13), and at fixed voltage excursion, Eq. (15); d, ν the population's relaxation rate and ringing frequency	–
e_0	half the maximal firing rate ($\sigma \to 2e_0$ as $v \to \infty$)	s^{-1}
v_0	sigmoid inflection (half-maximal) potential	mV
ρ	sigmoid slope parameter	mV^{-1}
v^*	operating point (field-free fixed point of v)	mV
$v = y_1 - y_2$	net pyramidal membrane potential (LFP proxy)	mV
PEIX	Population Excitation–Inhibition Index (a σ'-based brain-state index)	–
Single-column Jansen–Rit		
A, B	excitatory, inhibitory PSP gains	mV
a, b	excitatory, inhibitory synaptic rates (inverse time constants)	s^{-1}
y_1, y_2	excitatory, inhibitory post-synaptic potentials	mV
C, $C_1 - C_4$	intracolumn connectivity constants	–
p	external input rate; JR bifurcation parameter	s^{-1}
Applied TI field and coupling		
ε	field modulation amplitude, JR and LaNMM (a membrane polarization entering the sigmoid argument)	mV
A_f, $I_E(t)$	amplitude of the field's current-like forcing $I_E(t) = P\,E(t)$ in the E voltage equation, NMM2/QIF (dimensionless MPR units of v_E)	–
m	modulation depth	–
f_c, $\omega_c = 2\pi f_c$	carrier frequency	Hz, $rad\,s^{-1}$
f_1, f_2	the two applied carrier tones ($f_c = (f_1 + f_2)/2$)	Hz
Δf, $\Omega = 2\pi\Delta f$	envelope (beat) frequency, $\Delta f = f_2 - f_1$	Hz, $rad\,s^{-1}$
$E(t)$, $s(t)$	applied field; membrane perturbation reaching the nonlinearity ($s = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}$, Eq. (8))	$V\,m^{-1}$; mV
λ, λ_0^c, λ_f^c	field-to-membrane coupling: generic, DC (door c), and carrier-effective ($\lambda_f^c = \lambda_0^c \lvert H_c \rvert$)	mV / ($V\,m^{-1}$)
$\Lambda[\hat{E}]$	direct (non-synaptic) field term in the sigmoid argument, Eq. (3); equals $s(t)$ in the JR column	mV
$H_c(\omega)$, H_{gate}	access filter of door compartment c; Na_v activation-gate optional channel-kinetic filter	–

Symbol	Meaning	Units
$H_s(\omega)$	synaptic (second-order) JR synaptic transfer function, §2.5	– mV s
$H_m(\omega),\ \tau_m$	somatic effective membrane low-pass and its time constant	–; ms
$\tau_s,\ \tau_{\text{rel}}$	synaptic and population relaxation times in the adiabatic reduction	ms
s_{syn}	filtered firing rate in Eq. (6), distinct from field polarization $s(t)$	s^{-1}
τ	fast-element (node/AIS) time constant; also $\tau = 1/\gamma$ in $\mathcal{Q} = \pi f_0 \tau$ effective access time used in the filter comparisons	ms
Network dynamics, response, and bifurcation		
$\lambda_\pm = -\gamma \pm i\omega_0$	leading Jacobian eigenvalue pair	s^{-1}
γ	damping of the critical mode ($\propto$ distance to the Hopf)	s^{-1}
$\omega_0,\ f_0 = \omega_0/2\pi$	natural (Hopf) frequency	$\text{rad}\,\text{s}^{-1}$, Hz
$\mathcal{Q} = \omega_0/2\gamma = \pi f_0 \tau$ $\tau_{\text{rev}} = 1/\gamma$	network reverberation time	s
$\mathcal{Q} = \omega_0/2\gamma$	near-Hopf resonance quality factor ($\tau = 1/\gamma$ the reverberation time)	–
$G(\Omega)$	network resonance gain (Lorentzian, $\propto 1/\gamma$ at ω_0) response to a generated beat drive; $\propto 1/\gamma$ near resonance	– input/output dependent
D	beat-source coefficient for the injection convention in Eq. (21)	model-dependent
A_Ω	demodulated response: lock-in amplitude at Ω of v (mV) or, in §4.7 and the QIF network, of a rate	mV or s^{-1}
$Y_\Omega,\ Y_{\text{LF}}$	detector (pre-network) output at Ω and at baseband, §2.4	–
$\mathcal{M}[\cdot]$	peak-to-peak excursion of a periodic quantity, Eq. (18)	–
η_{det}	TI/tACS drive ratio of a memoryless transfer at matched peak polarization, App. D	–
δ	threshold offset of a step detector, App. D	mV
$\ell(\omega),\ \ell_0$	AC space constant and its DC value, Eq. (23)	μm
PPC	pairwise phase consistency of single-neuron spike phases at Δf [32]	–
A_{open}	same lock-in, open loop (detector only, no recurrence); A_Ω/A_{open} dimensionless	mV
$I,\ Q$	in-phase, quadrature lock-in components	mV
$C^*,\ J^*$	coupling at the Hopf (criticality)	–
Laminar model (LaNMM)		
$P_1,\ P_2$	deep (alpha) and superficial (gamma) pyramidal populations	–
PV, SS, SST	parvalbumin, stellate, somatostatin interneurons	–
Exact mean field (NMM2 / MPR) and QIF network		
$r_E,\ r_I$	population firing rates (E, I)	s^{-1}
$v_E,\ v_I$	mean membrane potentials (E, I), in dimensionless MPR units	–
$\eta_j,\ \bar{\eta}$	per-neuron excitability; mean excitability (= common drive, bifurcation param.)	–
Δ	Lorentzian half-width of η_j (excitability heterogeneity)	–
$\tau_E, \tau_I, \tau_a, \tau_g$	E, I membrane and AMPA, GABA synaptic time constants	ms
$C\ (\equiv J)$	global synaptic coupling (inter-population gain)	–
$A_{ee}, A_{ei}, A_{ie}, A_{ii}$	synaptic weight matrix $(1, 1, 1, 2)$	–
$s_E,\ s_I$	second-order synaptic activations; $\hat{K}_a, \hat{K}_g$ their operators	s^{-1}
$V_j,\ V_{\text{peak}}$	QIF neuron potential; spike/reset threshold	mV
N	neurons per population (QIF; $N = 8000$)	–
$A_\Omega/\langle r_E \rangle$	Δf modulation depth (envelope-locked timing metric)	–

Symbol	Meaning	Units

Table 4: Symbols, meanings, and (nominal) units. Reused glyphs are disambiguated by context; see the head note on A and C.

B Simulated models: equations and parameters

The models are written in the NMM1 transfer-function formalism of Eq. (3) [15]: synaptic operators $\hat{L}$ transduce firing rates into PSPs, the PSPs sum with any direct field term $\Lambda[\hat{E}]$ in the sigmoid argument, and the sigmoid returns the rate. The Jansen–Rit column is given here; the laminar model is specified in [13, 25] with its field drive in Eq. (26), and the NMM2 PING motif and its QIF network in Eq. (27) and §3.4.

Jansen–Rit column

Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, external input p, and the applied-field perturbation $s(t)$ (§2.2) entering the pyramidal sigmoid,

$$\ddot{y}_0 = Aa\,\sigma(y_1 - y_2 + s(t)) - 2a\dot{y}_0 - a^2 y_0, \tag{28}$$

$$\ddot{y}_1 = Aa\,[\,p + C_2\,\sigma(C_1 y_0)\,] - 2a\dot{y}_1 - a^2 y_1, \tag{29}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{30}$$

integrated as six first-order ODEs; the LFP proxy is $v = y_1 - y_2$ [16].

Table 5: Jansen–Rit parameters [16].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

C TI requires detection: the Stuart–Landau The cubic Hopf normal form amplifies but cannot demodulatelacks a quadratic mixer

Amplification and detection are separable, and the universal Hopf normal form makes the separation sharp. The Stuart–Landau equation [33] is

$$\dot{z} = (\alpha + i\omega_0)z - |z|^2 z + \mathcal{F}(t), \qquad \gamma \equiv -\alpha > 0 \ \text{(stable side)},$$

$$\dot{z} = (\alpha + i\omega_0)z - |z|^2 z + \mathcal{F}(t), \qquad \gamma = -\alpha > 0, \tag{31}$$

describes a complex mode z with damping γ and natural angular frequency ω_0, driven by the TI field $\mathcal{F}$. Expanding $z = \sum_k Z_k(t) e^{ik\omega_c t}$ in carrier harmonics with slowly varying envelopes, the field forces only $k = \pm 1$ and the cubic feeds index $k = l - p + q$. The odd-k subspace is therefore invariant and attracting: if every even Z_k vanishes, then for even k both sources vanish—$\mathcal{F}_k = 0$, and $l - p + q$ cannot be even with l, p, q odd—leaving $\dot{Z}_k = [\alpha + i(\omega_0 - k\omega_c)]Z_k$, which decays at rate $\gamma > 0$. So z lives only on odd carrier harmonics, to all orders in ε. The beat line is a *baseband* signal and would sit in $k = 0$; since $Z_0 \equiv 0$, the response at Δf is identically zero. An *even* nonlinearity instead produces a nonzero $k = 0$ term through $\cos^2 \omega_c t \to \frac{1}{2}$—the very line the odd cubic cannot make.Numerically applied input $\mathcal{F}$. For a weak input $\mathcal{F} = \varepsilon f(t)$ on the stable side, the perturbative response has the form $z = \varepsilon z_1 + \varepsilon^3 z_3 + \cdots$. There is no second-order source: the first-order response contains only the applied frequencies, and the cubic first contributes at order ε^3. Thus the normal form has a resonant linear susceptibility but lacks the quadratic difference-frequency generation of Eq. (10). This statement concerns the weak-input expansion; it is not an exclusion of every higher-order intermodulation or strong-drive response.

In the numerical control ($\alpha = -0.5$, $f_0 = 10$ Hz, $f_c = 100$ Hz, $\varepsilon = 0.5$), a square-law detector on the same field gives a Δf gives a beat-frequency lock-in amplitude of 2.5×10^{-1}, while $\mathrm{Re}\, z$ gives $\sim 2 \times 10^{-8}$approximately 2×10^{-8}, at the numerical floor. Driven instead Driving the same mode directly in band , the same equation shows the recovers the forced $1/\gamma$ growthtoward the Hopf.

The normal form thus responds to tACS but is blind to TI. Amplification is the model-independent near-Hopf susceptibility, shared with any normal-form resonator; detection is the model-specific even-order term the normal form lacks. A critical resonator alone cannot demodulate. . The control separates resonant amplification from the nonlinear source required to drive it at the beat.

D Efficiency of a memoryless transfer against a direct in-band drive

This appendix keeps the bound on the TI /tACS ratio for a bounded monotone *memoryless* transfer, at matched peak post-coupling polarizationand balanced tones. It is a statement about that comparator: it does not govern the dynamic detector of §2.3, whose source is a voltage-domain mixing term of a different dimension, and it is not a limit on For the logistic family of memoryless transfers, compare balanced TI tones of polarization amplitude ε each with a direct in-band drive of amplitude 2ε, matching peak polarization. Their leading drive amplitudes are $|\sigma''|\varepsilon^2/2$ and $2\sigma'\varepsilon$, so $\eta_{\text{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$. This ratio concerns the specified transfer family and input matching, not TI in general . Writing the TI drive $\frac{1}{2}\sigma''\varepsilon_1\varepsilon_2$ against the tACS drive $\sigma'\varepsilon$ at the same peak polarization, with $\varepsilon_1 = \varepsilon_2 = \varepsilon$, the ratio is $\eta_{\text{det}} = \sigma''\varepsilon/4\sigma'$, proportional to ε: a square-law detector gets less efficient as the signal weakensor the dynamic detector of §2.3.

The obvious escape is the operating point, since both σ'' and σ' depend on it and the column might sit somewhere far more favorable. It does not escape. For the logistic transfer, writing $s = \sigma/2e_0 \in (0,1)$, we have $\sigma' = 2e_0\rho\, s(1-s)$ and $\sigma'' = 2e_0\rho^2 s(1-s)(1-2s)$, so $\sigma''/\sigma' = \rho(1-2s)$ and, since PEIX $= 4s(1-s)$ Writing $s = \sigma/(2e_0) \in (0,1)$ gives $(1-2s)^2 = 1 - \text{PEIX}$,

$$\eta_{\text{det}} = \frac{\rho\,\varepsilon}{4}\sqrt{1-\text{PEIX}} \;\le\; \frac{\rho\,\varepsilon}{4}.$$

$\sigma' = 2e_0\rho s(1-s)$ and $\sigma'' = 2e_0\rho^2 s(1-s)(1-2s)$. Since PEIX $= 4s(1-s)$, the leading ratio satisfies

$$\eta_{\text{det}} \simeq \frac{\rho\,\varepsilon}{4}\sqrt{1-\text{PEIX}} \;\le\; \frac{\rho\,\varepsilon}{4}. \tag{32}$$

The efficiency is therefore state-dependent exactly as the mechanism predicts—maximal where the population is saturated and far from the inflection, and vanishing at PEIX $= 1$, where the transfer is locally linear and cannot demodulate at all—but at fixed ρ it is *bounded* by $\rho\varepsilon/4$ whatever the state. The maximum is one of It vanishes at the inflection, where the quadratic coefficient is zero, and approaches its bound in the saturated tails. This maximizes a ratio rather than of an absolute response: deep in the saturated tails σ' and σ'' both vanish and the TI and tACS drives disappear together. At the physiological Jansen–Rit slope (both drives vanish in those tails. For the JR slope $\rho = 0.56$ mV^{-1}, transfer width $1/\rho \approx 1.8$ mV) that ceiling the perturbative bound is 1.4% at Vieira et al.'s fields $\varepsilon = 0.1$ mV and 0.28% at human ones. Brain state moves the efficiency, but at fixed ρ it cannot lift the transfer off that ceiling. $\varepsilon = 0.02$ mV.

The slope is itself a parameter, so the question is what the whole sigmoid family can do. The logistic transfer depends on voltage only through $\rho(v - v_0)$, so writing

$$u = \rho\,\varepsilon, \qquad x = \rho\,(v^* - v_0), \tag{33}$$

we have $\sigma(v^* + \varepsilon q(t)) = 2e_0\, L(x + u\, q(t))$ for any normalized waveform q, with $L(z) = (1+e^{-z})^{-1}$. Every Fourier amplitude, and hence their ratio, is a function of (u, x) alone; the factor $2e_0$ cancels. Maximizing over v^* and ρ at fixed ε is therefore the same as maximizing over (u, x), and the family supremum is independent of field strength. Evaluating the exact cycle response rather than its quadratic truncation and optimizing x at each u, the ratio rises monotonically over the range explored and converges to the hard-threshold limit

$$\sup_{\rho,\, v^*} \eta_{\text{det}} = 0.3479, \tag{34}$$

approached as $\rho \to \infty$ at a threshold offset δ (the distance from the operating point to the inflection) with $\delta/2\varepsilon = 0.523$. There is no finite optimal transfer width: narrower is always better, and reaching 95% of the ceiling requires $u \gtrsim 6$, that is $1/\rho \lesssim \varepsilon/6$.

Two features of that limit decide the reading. Such a detector is not weak. At the optimum, and at the Vieira-scale polarization $\varepsilon \simeq 0.1$ mV, its open-loop beat drive exceeds the physiological-slope sigmoid's by some three orders of magnitude for the same e_0 and waveform, because the field excursion is about twice the threshold offset and the transfer switches across its full range instead of responding linearly. The disparity grows as the field weakens, the smooth sigmoid's response

falling as ε^2 while a threshold repositioned with the field keeps its excursion. The obstacle is the width. At Vieira et al.'s fields the ceiling is approached only for $1/\rho \lesssim 17$ μV, and at human polarizations for $1/\rho \lesssim 3$ μV, with the threshold offset held to the same order as the applied polarization. The width of a neural-mass sigmoid is set by the spread of thresholds and noise across $\sim 10^4$ neurons and is of order millivolts. A microvolt width lies far below that, and reads more naturally as an idealized microscopic threshold or rectifier than as a neural-mass sigmoid; we do not offer it as a measured single-cell property, real thresholds being themselves noisy and state-dependent. The regime in which the family approaches its ceiling is thus one in which the "population sigmoid" has ceased to describe a population.

The limit is a step detector, not an envelope follower. For a unit step at offset δ the carrier-averaged transduction is $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and zero below; with $\beta = \delta/2\varepsilon$ its tACS fundamental is $(2/\pi)\sqrt{1-\beta^2}$ and its TI coefficient reduces by parts to $\eta_{\text{det}} = 2\beta\,[K(m) - E(m)]/(\pi\sqrt{1-\beta^2})$ with $m = 1-\beta^2$ and K, E the complete elliptic integrals, giving 0.3479 at $\beta = 0.523$ by direct quadrature and by the elliptic form alike (`threshold_bound_check.py`). At that optimum the step's second-to-first harmonic magnitude ratio is 0.151, not the 1/5 of a linear envelope follower. A bounded nondecreasing transfer is a positive mixture of shifted steps and both Fourier coefficients are linear in the transfer, so the step optimum bounds the family; the maximum itself is a numerical evaluation, not a certified bound.

E Extension: the laminar column at low carriers

This extension drives the two-band laminar column (LaNMM, §2.1) with carriers of 25–80 Hz and polarizations up to 3 mV, outside the perturbative regime of the main text and below the carriers TI uses. It is included for what it shows about a band-limited intermediary between field and slow generator; it does not carry the kHz argument.

The single JR column isolates and lets us analyze the mechanism; the laminar model (LaNMM, §2.1) confirms it in a cortical column with its *own* two resonances. Driving one population with the two-tone field of Eq. (26) and reading the alpha-band power of the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials (Methods, Fig. S2), we recover the same envelope resonance, now carrying the column's intrinsic rhythms.

As a consistency check, driving the deep P_1 (alpha) loop directly reproduces the alpha-band beat resonance of the JR column on both the stable-focus and limit-cycle sides, with the sweep and its code in the repository; what the laminar model adds is the carrier-dependent structure below.

One qualification governs the laminar sweeps. They run to $\varepsilon = 3$ mV on P_2, above the sigmoid voltage scale $1/\rho \approx 1.8$ mV, so they sit deliberately in a nonlinear, non-perturbative regime. They are a demonstration of tongues and frequency capture, not a verification of the weak-field square law; that verification is the low-amplitude JR work (Fig. 6b), which fits the square law at 0.05–0.2 mV, one to two orders of magnitude below $1/\rho$.

Sweeping the field amplitude ε and beat frequency Δf at a fixed carrier produces *nonlinear Arnold tongues*: alpha power in P_1 is maximal when Δf matches the deep alpha (∼10 Hz) and grows with ε (Figs. S1a,b and S7a,b), envelope resonance and phase locking of the slow generator to the recovered beat component, as in the JR resonance curves but now in a two-band column. Outside the tongue the drive *reduces* alpha-band power: with the field on P_1, 95% of the amplitude–beat grid and 97% of the carrier–beat grid lie below the field-free baseline (Fig. S7), the drive pulling the autonomous alpha out of the 8–12 Hz band or suppressing it; with the field on P_2 the fractions are 44% and 31% (Fig. S1). The earlier version of these maps showed only the positive part of the change.

The carrier dependence illustrates the two coupling limits of §2.2–§2.6 within this model's 25–80 Hz carrier range. Sweeping carrier f_c against beat Δf: when the field is delivered to the *superficial* loop (P_2), strong P_1 demodulation requires a *double* resonance—f_c near the P_2 gamma (≈44 Hz) *and* Δf near the P_1 alpha (≈10 Hz) (Fig. S1c,d): the fast loop acts as a tuned radio-frequency (RF) front end that must itself be excited before it can pass a demodulated envelope to P_1, and off that gamma resonance the response collapses to ∼7% of its peak. When the field is delivered *directly* to the slow generator (P_1), the picture inverts: the envelope-locked response persists at every carrier from 25 to 80 Hz, falling only to ∼25% of its peak, and the ridge sits at $\Delta f \approx 10.5$ Hz even for a 60 Hz carrier (Fig. S7c)—broadband , broadband in f_c, as Eq. (10) predicts for direct coupling. This contrast requires the phase-sensitive read-out: because P_1 free-runs at its own alpha, band power alone buries a weak envelope-locked drive under an autonomous rhythm that is not phase-locked to the beat, and only the lock-in at Δf separates the two (Methods). TI's kHz carriers lie outside this model's range; what the contrast shows is that a carrier matched to a fast (gamma) network resonance couples best, while the envelope response persists, weaker, off resonance. This both confirms the JR mechanism and refines it: the "inert carrier" of §2.5 is the direct-coupling idealization, while a band-limited intermediary makes the carrier choice matter, consistent with frequency-tagging/intermodulation paradigms [60].

Concretely the LaNMM predicts that TI efficacy should (i) peak when Δf matches a region's intrinsic alpha; (ii) widen and strengthen with modulation amplitude (the tongue); and (iii) for fields delivered via superficial fast circuits, be largest when the carrier is matched to a gamma network resonance—e.g. resonance, for example two nearby gamma carriers summing to a ∼40 Hz-

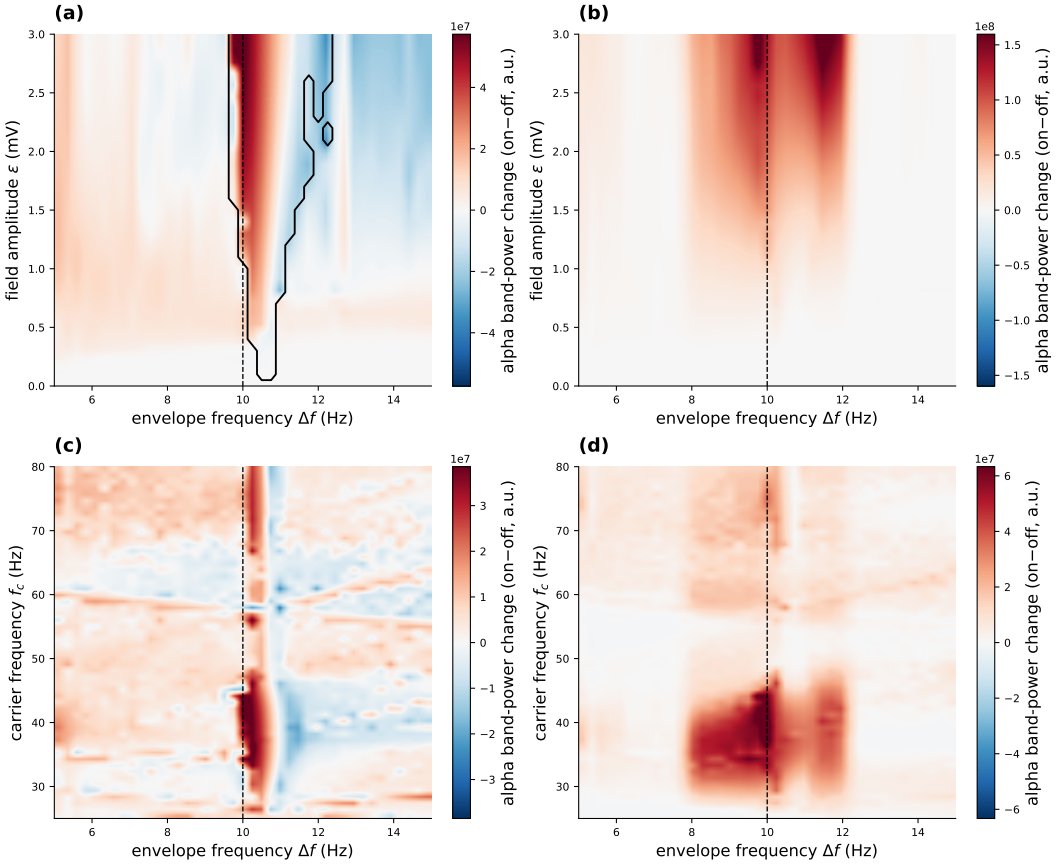


Figure S1: LaNMM Arnold tongues: two-tone TI field on the superficial P_2 (gamma) loop. The field enters the P_2 sigmoid argument as a membrane polarization ε (mV) with all external drives flat. Color shows the signed change of alpha-band (8–12 Hz) power from the unstimulated $\varepsilon = 0$ baseline, on a zero-centered scale (arbitrary units). *(a,b)* P_1 and P_2 enhancement vs. ε and Δf (carrier 40 Hz), peaking at the P_1 alpha (10.0 Hz, dashed). The black contour in *(a)* is the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), drawn because band power alone cannot separate a driven from an autonomous rhythm; it is a dominance screen, not a phase-capture test. The $\varepsilon = 0$ row is excluded, since with no drive the autonomous 10.67 Hz alpha already lies within the tolerance of two grid points (a resolution floor of 0.5 Hz); the set widens to 2.75 Hz at 3 mV. *(c,d)* Enhancement vs. carrier and Δf at $\varepsilon = 1.5$ mV: P_1 demodulation needs *both* a carrier near the P_2 gamma (≈ 44 Hz) and Δf near the P_1 alpha—a double resonance.

centered AM field—while field, while remaining present, weaker, for off-resonance carriers. The third is untested: existing TI studies of deep targets sweep the *beat* frequency rather than the carrier against a network resonance [61, 62], so they neither support nor contradict it.

F Vocabulary and metrics

The three terms (forced response, entrainment, gating) are defined in Methods; this appendix gives the metrics behind them. *(i) Forced response.* Below the Hopf the column does not oscillate on its own; the field elicits a driven response whose susceptibility we quantify by the lock-in amplitude of the read-out at the beat Δf (Eq. (25)), and there is no autonomous rhythm to entrain (J-curve, timing, tACS). *(ii) Frequency entrainment.* Above the Hopf the column free-runs at f_0; genuine entrainment means its dominant frequency is *pulled* to Δf. We screen for it with the dominant output frequency f_{out} (largest non-DC FFT peak) and the criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz, mapping the tongue and locking range (Fig. S4); f_{out} is used rather than a phase-locking value [63] because, when an autonomous rhythm and a driven component coexist, instantaneous-phase estimates conflate them. The dominance criterion is a screen, not a capture test: the largest peak can move to the drive frequency because a driven component outgrows a persisting autonomous one. A capture test would follow the unwrapped phase difference and require the absence of secular drift; the maps therefore label the set the frequency-dominance set. *(iii) Envelope locking / gating.* A periodic drive can instead modulate the *amplitude* of a persisting rhythm at the beat without capturing its frequency—phase–amplitude couplingfrequency (phase–amplitude coupling) [34, 35]. We quantify this by the Δf-locked component of the population rate (lock-in of r_E at Δf, equivalently the rate *folded* on the Δf beat phase—averaged phase, averaged over many beat cycles as a function of phase within a single cycle, so that everything not locked to Δf averages away), which isolates the driven envelope from the intrinsic rhythm (the above-Hopf QIF panel, with its ∼11 Hz detuning, is a case of gating, not entrainment).

The Arnold-tongue protocol for Fig. S4 places the column on the limit-cycle side (input p between the SNIC and the Hopf, free-running at $f_0(p)$) and drives it with $s(t) = \varepsilon \cos \Omega t$; the implementation, including the grid widths and the clipping check, is in the released code. One subtlety: as onset is approached the cycle amplitude vanishes and the tongue merges into the forced-resonance bandwidth, so the supercritical locking range and the subcritical $1/\gamma$ susceptibility coincide there by construction—which is why responsiveness peaks at criticality on both sides.

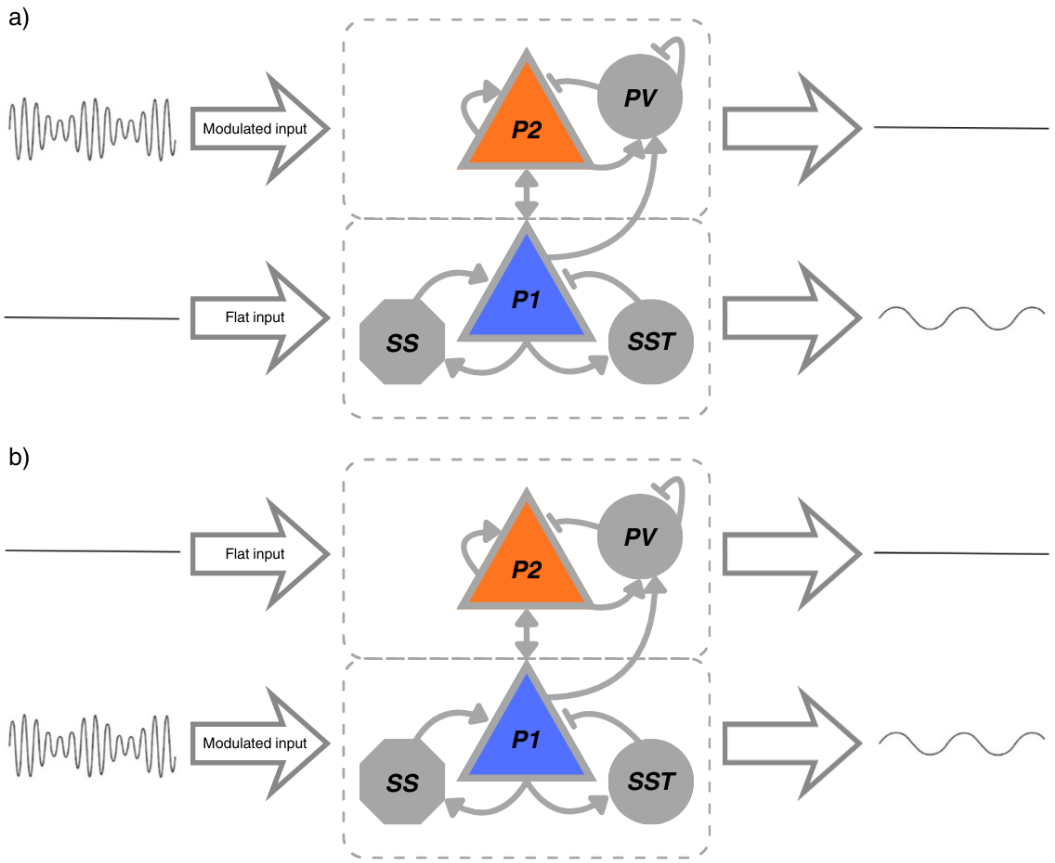


Figure S2: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the TI field drives P_2 while P_1 receives a flat drive; *(b)* the TI field drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the generated beat-frequency component is transferred to the deep loop: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency component (right traces).

G Supplementary figures

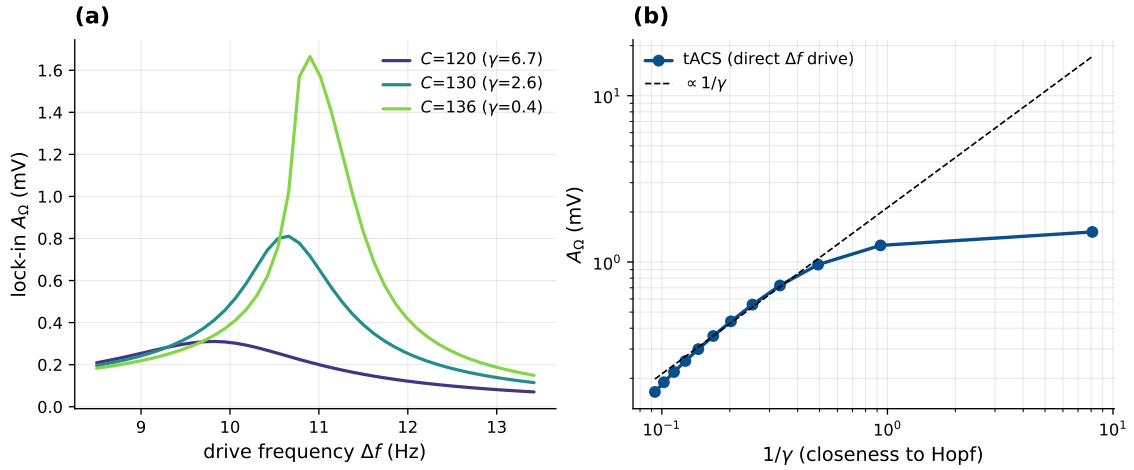


Figure S3: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the operating point pinned, sweeping the coupling $C\,(=J)$ toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 9), but linear in the field rather than square-law. Throughout, the column is a stable focus ($\gamma > 0$).

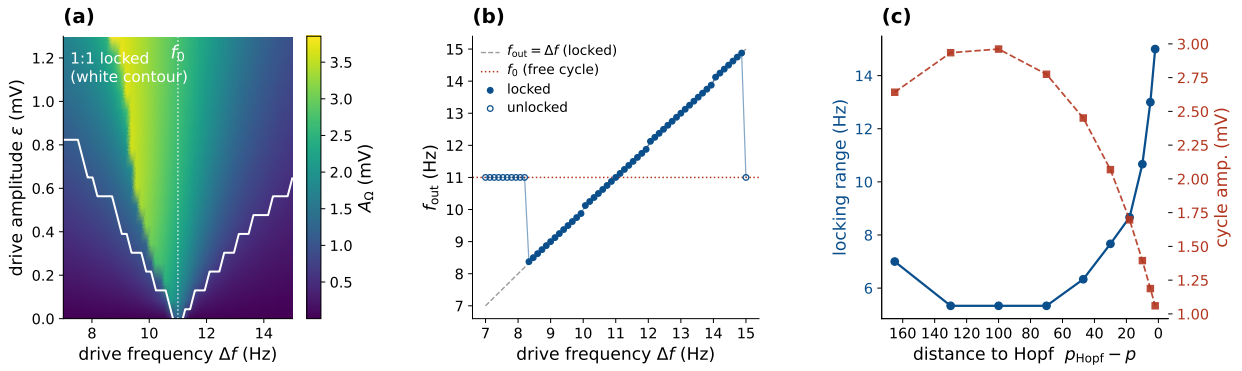


Figure S4: Above the Hopf: entrainment, not $1/\gamma$ amplification (Jansen–Rit). The column is set in its autonomous alpha limit cycle ($p = 250$, $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid. *(a)* The 1:1 Arnold tongue—where tongue, where the dominant output frequency locks to Δf—is , is a wedge anchored at f_0 that widens with drive. *(b)* Inside the tongue the output follows Δf; outside, it falls back to f_0. *(c)* Toward the Hopf at fixed drive the locking range grows as the autonomous cycle amplitude vanishes (blue, left axis; red dashed, right axis). Above the Hopf amplification reads as tongue width rather than as the diverging $1/\gamma$ gain of the stable-focus side (Fig. 9): the column is most responsive near criticality on both sides.

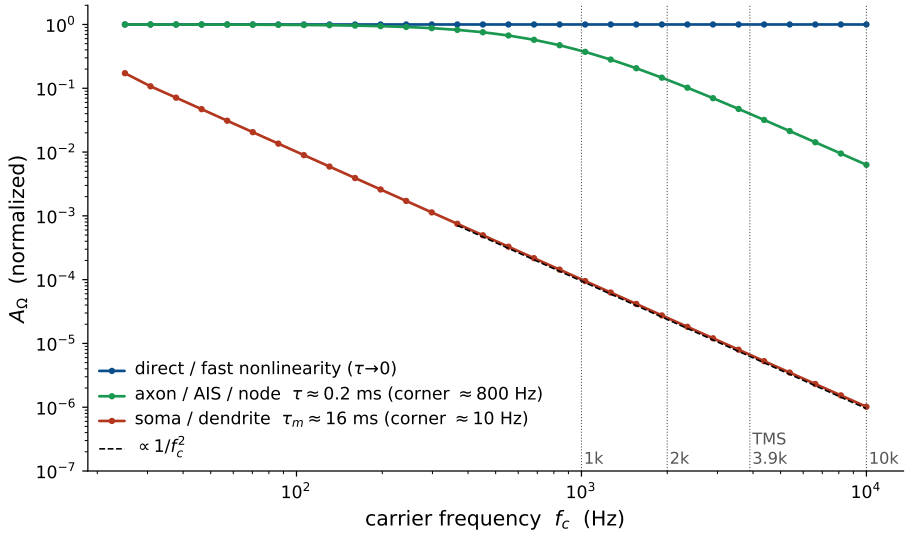


Figure S5: **Somatic demodulation dies at kHz, but a fast element survives (Jansen–Rit, open-loop detector). Carrier attenuation depends on the assumed access time (Jansen–Rit, open loop).** Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide), killing kHz demodulation; a the assumed fast element response (axon/AIS/node, $\tau \approx 0.2\tau = 0.2$ ms) retains an appreciable response into about 0.14 of the kHz band (∼0.14 direct response at 22 kHz , ∼0.04 and 0.04 at the TMS peak 3.9 kHz). The same membrane filter attenuates TMS at the somaThese are sensitivity cases, which is why neither TMS nor TI should be thought not measurements of as driving the soma at kHzanatomical compartments.

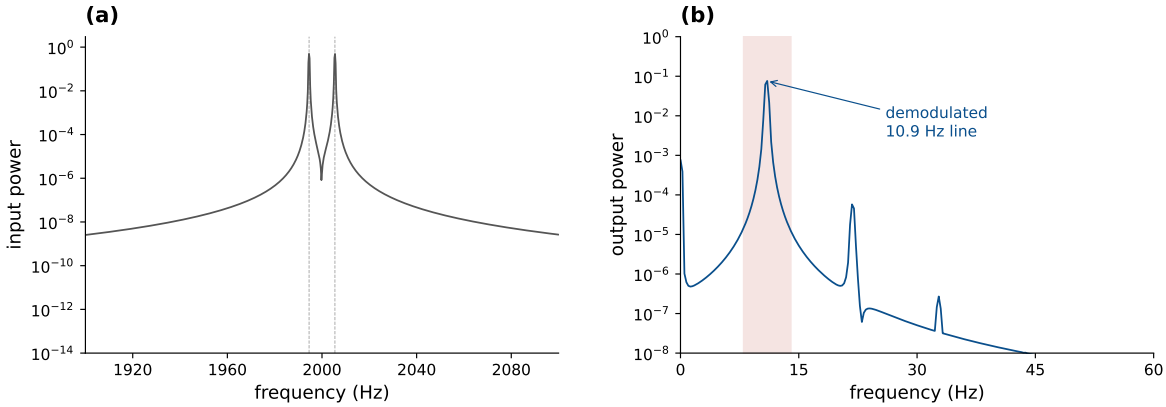


Figure S6: **Genuine two-tone TI at a real kHz carrier (Jansen–Rit, fast-element route). Two-tone TI at a kHz carrier (Jansen–Rit, assumed fast access).** Two equal tones at $f_1 = 1994.54$ and $f_2 = 2005.46$ Hz, beating at 10.91approximately 10.9 Hz, pass through the fast-element assumed low-pass ($\tau = 0.2$ ms) into the recurrent column. *(a)* The input amplitude spectrum has lines only at f_1 and f_2; in the 8–14 Hz band it is 2.2×10^{-13}, zero to numerical precision. *(b)* The output carries a demodulated line at the beat, 10.9 Hz.

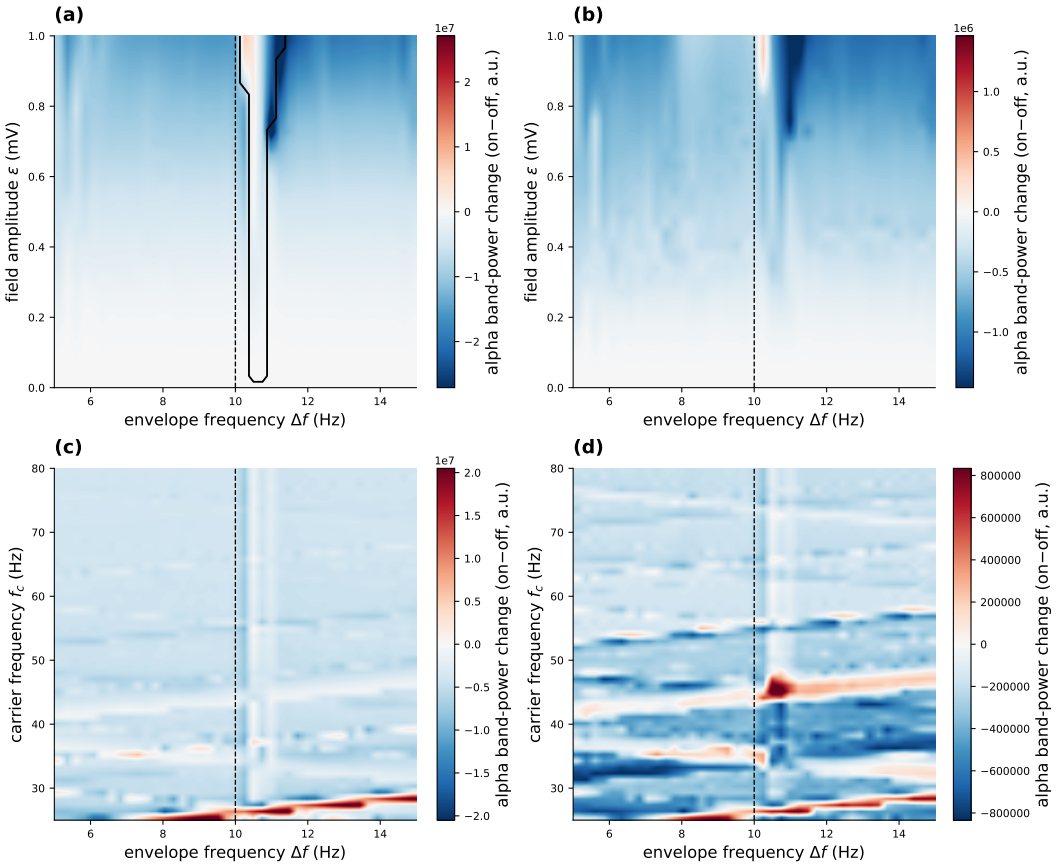


Figure S7: LaNMM Arnold tongues: two-tone TI field delivered directly to the deep P_1 (alpha) loop. As Fig. S1 but with the field on P_1. The deep loop is captured by a much weaker field than the superficial one, so the amplitude range is $\varepsilon \leq 1$ mV here against 3 mV there; above ≈ 2 mV the drive captures essentially the whole Δf sweep and leaves the perturbative regime the theory describes. Color: signed change of alpha-band power from the $\varepsilon = 0$ baseline, zero-centered; most of the grid is suppression. *(a,b)* P_1 and P_2 enhancement vs. ε and Δf; the black contour in *(a)* outlines the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), computed from the dominant output frequency because band power alone cannot separate a driven from an autonomous rhythm; the $\varepsilon = 0$ row is excluded (resolution floor 0.5 Hz). The set widens to 1.25 Hz at $\varepsilon = 1$ mV. *(c,d)* Enhancement vs. carrier frequency and Δf at $\varepsilon = 0.5$ mV. P_1 responds across a *broad* carrier band—a vertical ridge at $\Delta f \approx 10$ Hz, the direct-coupling limit of Eq. (10). Read with the lock-in at Δf the response is retained at every carrier from 25 to 80 Hz (median 25% of peak), in contrast to the $\sim$7% collapse off resonance under superficial drive.

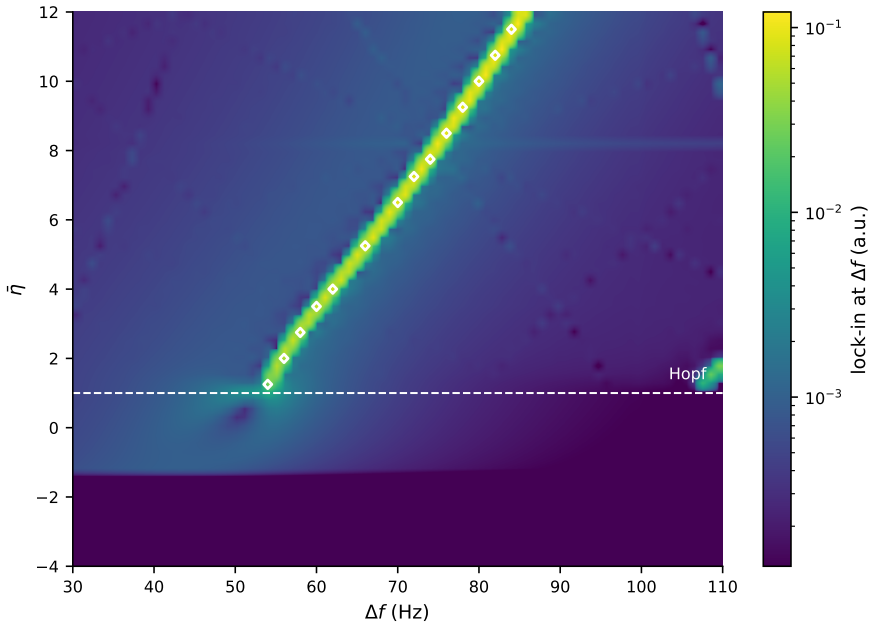


Figure S8: NMM2 PING resonance map (exact mean field). Demodulated response over the (Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency, and its diagonal slope—with f_0 rising with $\bar{\eta}$ —reflects the state-dependent transfer of NMM2, absent in the static-sigmoid models for these parameters. White markers outline the 1:1 captured set (Appendix F), drawn only above the Hopf (dashed line) where there is an autonomous rhythm to capture; below it the same color field is a forced response. Lock-in amplitude cannot separate the two, since both place power at Δf.

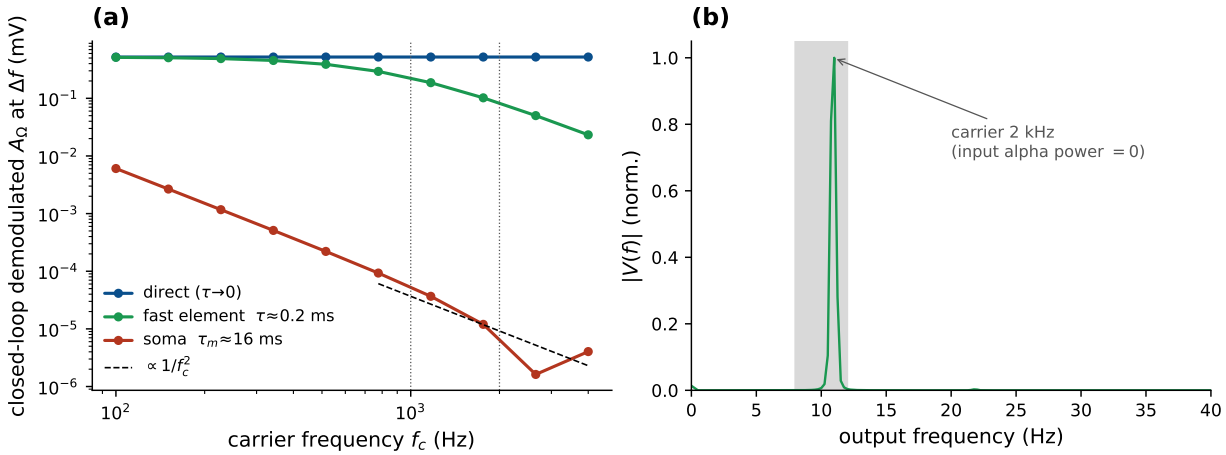


Figure S9: Demodulation at a genuine kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three coupling routesaccess models: direct ($\tau \to 0$, carrier-independent), fast element (axon/AIS/node, $\tau \approx 0.2\tau = 0.2$ ms; survives into the kHz band), and soma somatic ($\tau_m \approx 16\tau_m = 16$ ms; collapses as $1/f_c^2$ roll-off). The fast time is assumed rather than assigned to a measured compartment. *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: a clean alpha line at f_0, synthesized by the column although the input carries no alpha power.

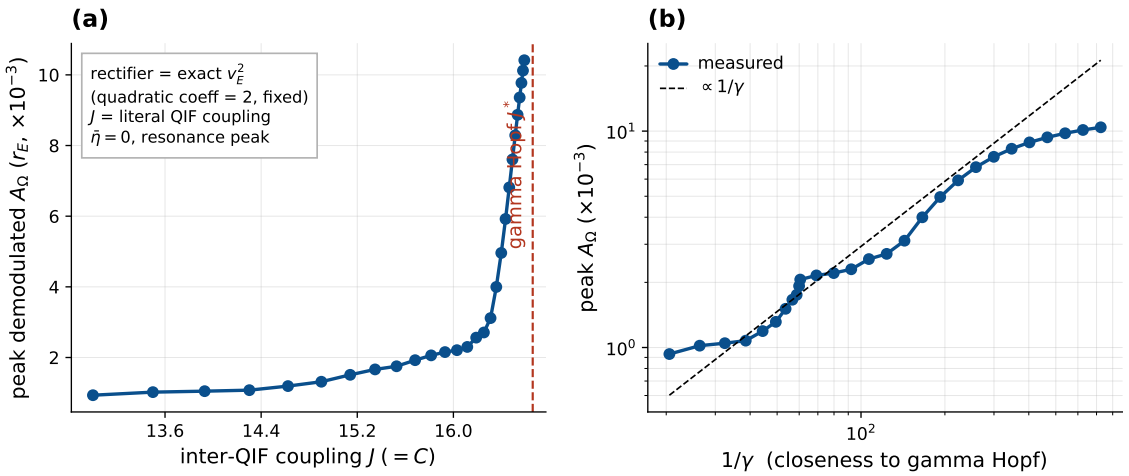


Figure S10: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). $J\,(=C)$ is the literal inter-QIF synaptic weight; at this carrier the generated source is operating-point independent (§2.3), so no pinning is needed. *(a)* The peak demodulated response of r_E grows as $J \to J^*$ (gamma Hopf, $\bar{\eta} = 0$; stable focus, $\gamma > 0$). *(b)* The gain grows as $1/\gamma$ (dashed), saturating near criticality—the JR law, now with coupling and rectifier both derived.

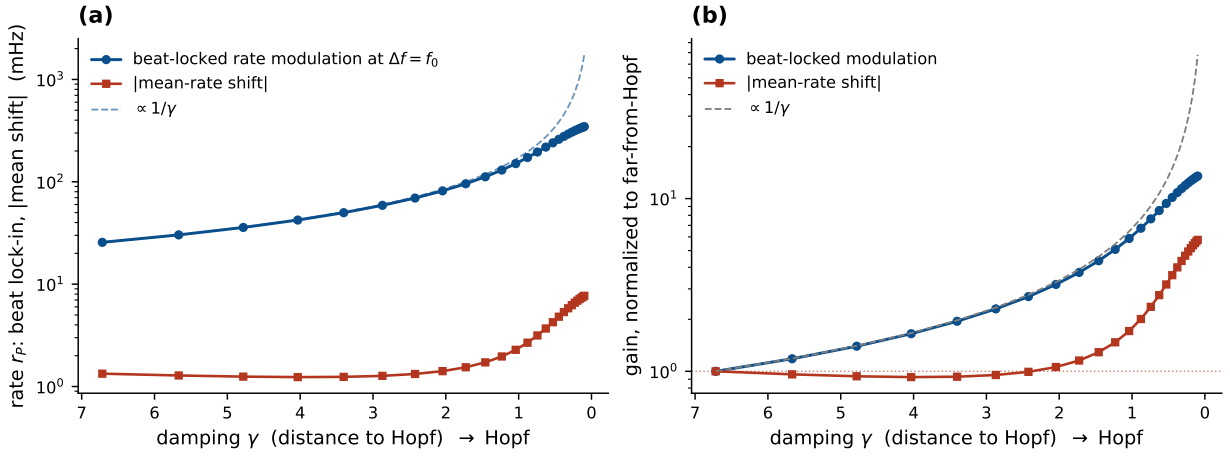


Figure S11: Envelope-locked modulation against mean-rate shift (Jansen–Rit, detection pinned). As the column is driven toward the Hopf (from the stable side, $\gamma > 0$) through the coupling at fixed $\sigma''(v^*)$, both quantities are read from the pyramidal rate $r_P = \sigma(v + s)$, in mHz. *(a)* The beat-locked modulation at $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed) before saturating; the magnitude of the mean-rate shift (red) stays at a few mHz and rises near the Hopf. *(b)* Both normalized to their far-from-Hopf values (log scale): the modulation gain follows $1/\gamma$; the mean-shift gain rises later and less. Modulation exceeds mean shift by 19× far from the Hopf and 45–66× near it.

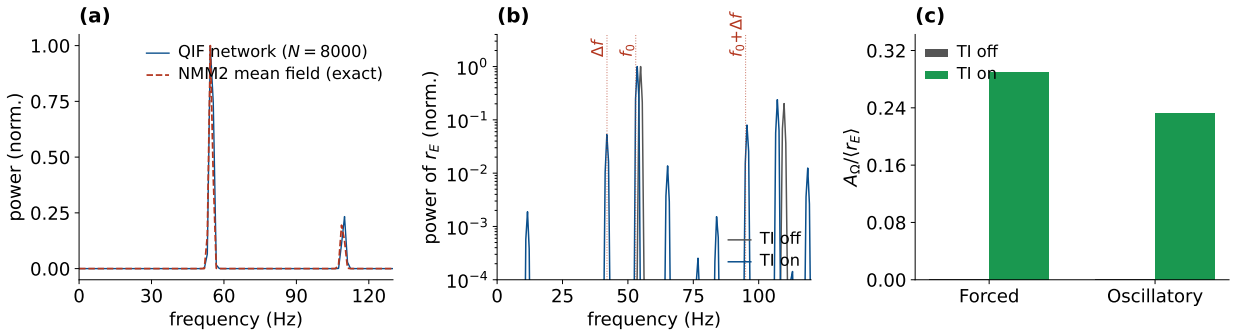


Figure S12: Timing, not rate, in the QIF network. *(a)* Field-free power spectrum of $r_E(t)$, normalized: the spiking network ($N = 8000$) matches the exact NMM2 mean field in gamma frequency (∼53 Hz); a normalized spectrum does not test the mean rate. *(b)* Intermodulation above the Hopf (TI off vs. on, log power): the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the intrinsic 53 Hz gamma, raising the direct Δf line and the sum sideband near 95 Hz, both absent with TI off. This is the sideband signature of gating rather than entrainment. *(c)* The Δf modulation depth $A_\Omega/\langle r_E \rangle$, off vs. on: below 0.003 rising to 0.29 (forced) and 0.23 (oscillatory), while the mean rate rises by 8–10%. Modulation depth is reported rather than the off→on lock-in ratio, whose denominator is the field-free floor.